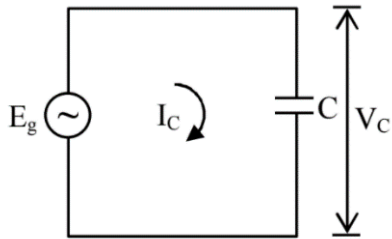
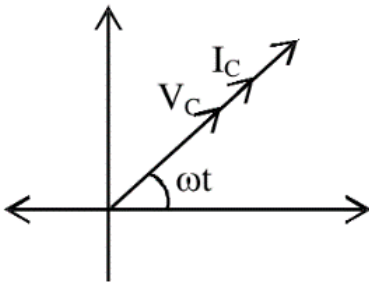


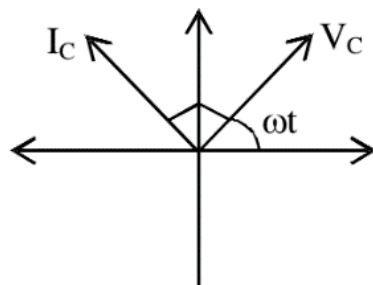
Q1: In a circuit consisting of a capacitance and a generator with alternating emf $E_g = E_{g0} \sin \omega t$, V_C and I_C are the voltage and current. Correct phasor diagram for such circuit is:



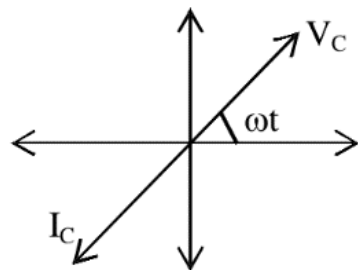
(A)



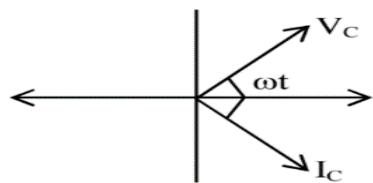
(B)



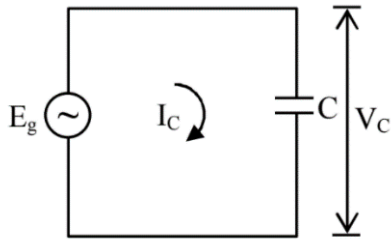
(C)



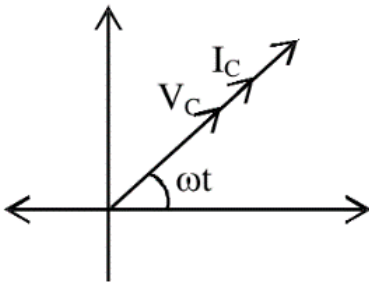
(D)



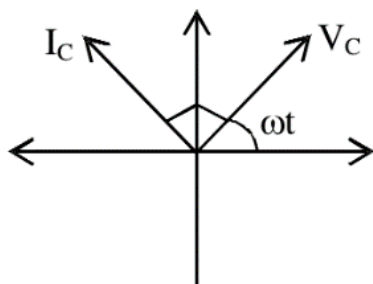
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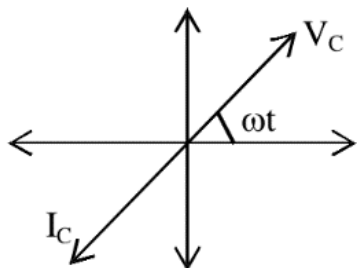
(A)



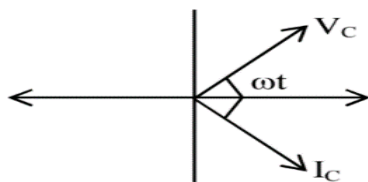
(B)



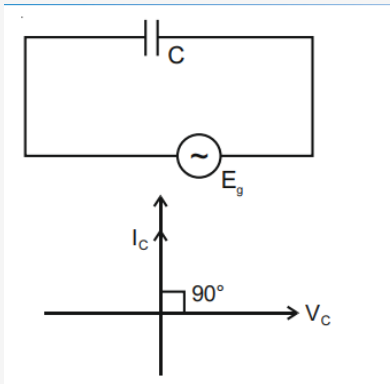
(C)



(D)



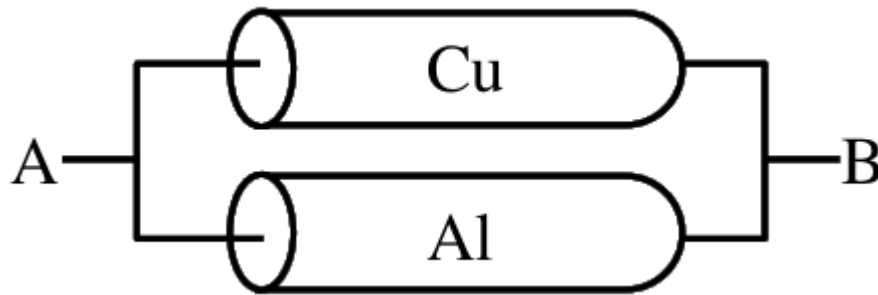
Solution:



and phasors rotate by ωt .

Q2: A Copper (Cu) rod of length 25 cm and cross-sectional area 3 mm^2 is joined with a similar Aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B.

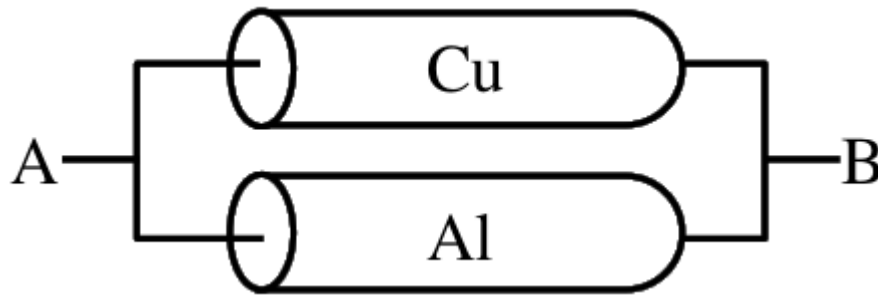
(Take Resistivity of Copper = $1.7 \times 10^{-8} \Omega m$ Resistivity of Aluminium = $2.6 \times 10^{-8} \Omega m$)



- (A) $2.170 \text{ m}\Omega$
- (B) $1.420 \text{ m}\Omega$
- (C) $0.0858 \text{ m}\Omega$
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Solution:

$$R_{Cu} = \frac{\rho_{Cu} \times \ell}{A}$$

$$R_{Al} = \frac{\rho_{Al} \times \ell}{A}$$

$$R_{Eq} = \frac{\rho_{Cu} \times \rho_{Al}}{\rho_{Cu} + \rho_{Al}} \times \left(\frac{\ell}{A} \right)$$

$$\begin{aligned} &= \frac{1.7 \times 10^{-8} \times 2.6 \times 10^{-8}}{(1.7 + 2.6) \times 10^{-8}} \times \frac{0.25}{3 \times 10^{-6}} \\ &= 0.858 \, m\Omega \end{aligned}$$

Q3: What will be the projection of vector $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ on vector $\vec{B} = \hat{i} + \hat{j}$?

(A) $\sqrt{2}(\hat{i} + \hat{j} + \hat{k})$

(B) $2(\hat{i} + \hat{j} + \hat{k})$

(C) $\sqrt{2}(\hat{i} + \hat{j})$

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(D) $(\hat{i} + \hat{j})$

Solution:

$$\begin{aligned}\text{Projection} &= \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} (\hat{B}) \\ &= \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} + \hat{j})}{\sqrt{2}} \frac{(\hat{i} + \hat{j})}{\sqrt{2}} \\ &= \frac{2}{\sqrt{2}} \frac{(\hat{i} + \hat{j})}{\sqrt{2}} \\ &= (\hat{i} + \hat{j})\end{aligned}$$

Q4: A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the work done by the porter in lowering the suitcase. (*take $g = 9.8 \text{ ms}^{-2}$*)

- (A) $- 62720.0 \text{ J}$
- (B) $- 627.2 \text{ J}$
- (C) $+ 627.2 \text{ J}$
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(A) -62720.0 J

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(D) 784.0 J

Solution:

$$\begin{aligned} W &= -N \times \Delta x \\ &= -80 \times 9.8 \times \frac{80}{100} \\ &= -627.2 \text{ J} \end{aligned}$$

Q5: T_0 is the time period of a simple pendulum at a place. If the length of the pendulum is reduced to $\frac{1}{16}$ times of its initial value, the modified time period is:

- (A) T_0
- (B) $8\pi T_0$
- (C) $4T_0$
- (D) $\frac{1}{4}T_0$

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Solution:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T' = \frac{T_0}{4}$$

Q6: Statement I: The ferromagnetic property depends on temperature. At high temperature, ferromagnet becomes paramagnet.

Statement II: At high temperature, the domain wall area of a ferromagnetic substance increases.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are true
- (C) Both Statement I and Statement II are false
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Solution:

With increase in temperature domain volume decreases.

Statement 1 true

Statement 2 false

Q7: A bullet of ' $4g$ ' mass is fired from a gun of mass 4 kg . if the bullet moves with the muzzle speed of 50 ms^{-1} , the impulse imparted to the gun and velocity of recoil of gun are:

- (A) $0.4\text{ kg ms}^{-1}, 0.1\text{ ms}^{-1}$
- (B) $0.2\text{ kg ms}^{-1}, 0.05\text{ ms}^{-1}$
- (C) $0.2\text{ kg ms}^{-1}, 0.1\text{ ms}^{-1}$
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Solution:

$$m_{\text{Bullet}} = 4\text{ g}, M_{\text{Gun}} = 4\text{ kg}$$

$$v_{\text{Bullet}} = 50\text{ m/s}$$

$$\text{Now } P_B = P_g$$

$$P_g = m \times v_{\text{Bullet}}$$

$$= \frac{4}{1000} \times 50$$

$$= 0.2\text{ kg m/s}$$

$$\text{So impulse} = 0.2\text{ kg m/s}$$

$$v_G = \frac{0.2}{M_{\text{Gun}}} = \frac{0.2}{4} = 0.05\text{ m/s}$$

Q8: Choose the correct option:

- (A) True dip is not mathematically related to apparent dip.
- (B) True dip is less than apparent dip.
- (C) True dip is always greater than the apparent dip.
- (D) True dip is always equal to apparent dip.

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Solution:

$$\tan(\theta_a) = \frac{\tan(\theta_T)}{\cos \phi}$$

$$\Rightarrow \theta_a \geq \theta_T$$

$\therefore$ True dip is less than apparent dip.

Q9: Consider a situation in which a ring, a solid cylinder and a solid sphere roll down on the same inclined plane without slipping. Assume that they start rolling from rest and having identical diameter.

The correct statement for this situation is:-

- (A) The sphere has the greatest and the ring has the least velocity of the centre of mass at the bottom of the inclined plane.
- (B) The ring has the greatest and the cylinder has the least velocity of the centre of mass at the bottom of the inclined plane.
- (C) All of them will have same velocity
- (D) The cylinder has the greatest and the sphere has the least velocity of the centre of mass at the bottom of the inclined plane.

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- (C) All of them will have same velocity
- (D) The cylinder has the greatest and the sphere has the least velocity of the centre of mass at the bottom of the inclined plane.

Solution:

$$\frac{K_T}{K_R} = \frac{MR^2}{I_{CM}}$$

I_{CM} is maximum for ring.

$\Rightarrow v$ is least for ring.

Q10: Consider a situation in which reverse biased current of a particular P – N junction increases when it is exposed to a light of wavelength $\leq 621 \text{ nm}$. During this process, enhancement in carrier concentration takes place due to generation of hole – electron pairs. The value of band gap is nearly.

- (A) 2 eV
- (B) 4 eV
- (C) 1 eV
- (D) 0.5 eV

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- (D) 0.5 eV

Solution:

$$\text{Band gap} = \frac{hc}{\lambda} = 2 \text{ eV}$$

Q11: A nucleus with mass number 184 initially at rest emits an α — particle. If the Q value of the reaction is 5.5 MeV, calculate the kinetic energy of the α — particle.

- (A) 5.0 MeV
- (B) 5.5 MeV
- (C) 0.12 MeV
- (D) 5.38 MeV

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Solution:

$$k_{\alpha} + k_N = 5.5$$

$$k = \frac{p^2}{2m}$$

$$\Rightarrow \frac{k_{\alpha}}{k_N} = \frac{180}{4} = 45 \Rightarrow k_{\alpha} = \frac{45}{46} \times 5.5 \text{ MeV}$$
$$= 5.38 \text{ MeV}$$

Q12: An electron of mass m_e and a proton of mass m_p are accelerated through the same potential difference. The ratio of the de – Broglie wavelength associated with the electron to that with the proton is:-

(A) $\frac{m_p}{m_e}$

(B) 1

(C) $\sqrt{\frac{m_p}{m_e}}$

(D) $\frac{m_e}{m_p}$

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Solution:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2km}}$$

$$\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{k_p m_p}{k_e m_e}} = \sqrt{\frac{m_p}{m_e}}$$

Q13: Match List – I with List – II

	List – I		List – II
(a)	$\omega L > \frac{1}{\omega C}$	(i)	Current is in phase with emf
(b)	$\omega L = \frac{1}{\omega C}$	(ii)	Current lags behind the applied emf
(c)	$\omega L < \frac{1}{\omega C}$	(iii)	Maximum current occurs
(d)	Resonant frequency	(iv)	Current leads the emf

Choose the correct answer from the options given below:

- (A) (a) – (ii) ; (b) – (i) ; (c) – (iv) ; (d) – (iii)
- (B) (a) – (ii) ; (b) – (i) ; (c) – (iii) ; (d) – (iv)
- (C) (a) – (iii) ; (b) – (i) ; (c) – (iv) ; (d) – (ii)
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(D) (a) – (iv) ; (b) – (iii) ; (c) – (ii) ; (d) – (i)

Solution:

$$\omega L = \frac{1}{\omega C}, X_L = X_C$$

So current in phase with EMF

At resonance, current have maximum value,

Q14: What should be the height of transmitting antenna and the population covered if the television telecast is to cover a radius of 150 km? the average population density around the tower is $2000/km^2$ and the value of $R_e = 6.5 \times 10^6 m$.

- (A) Height = 1731 m Population Covered = 1413×10^5
- (B) Height = 1241 m Population Covered = 7×10^5
- (C) Height = 1600 m Population Covered = 2×10^5
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Solution:

$$d = \sqrt{2R_e h}$$

$$\text{Area covered} = \pi d^2$$

$$h = \frac{d^2}{2R_e} = 1731 \text{ m}$$

Q15: What will be the average value of energy for a monoatomic gas in thermal equilibrium at temperature T ?

(A) $\frac{2}{3}k_B T$

(B) $k_B T$

(C) $\frac{3}{2}k_B T$

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Solution:

$$E = \frac{3}{2}K_B T$$

Q16: Intensity of sunlight is observed as 0.092 Wm^{-2} at a point in free space. What will be the peak value of magnetic field at that point? ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

- (A) $2.77 \times 10^{-8} \text{ T}$
- (B) $1.96 \times 10^{-8} \text{ T}$
- (C) 8.31 T
- (D) 5.88 T

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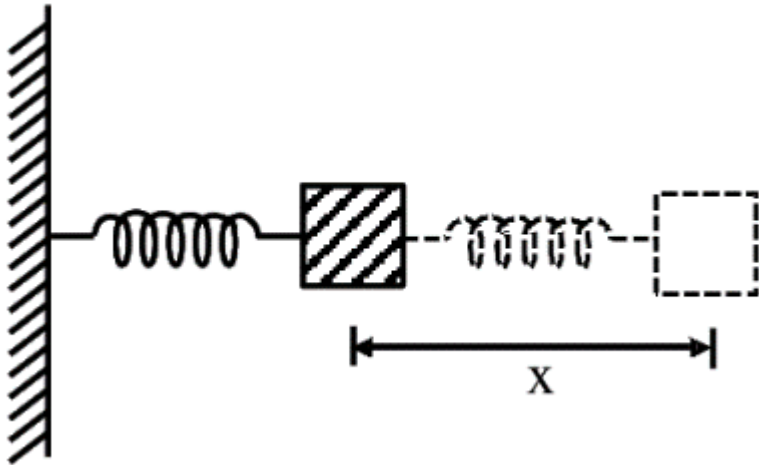
(C) 8.31 T

(D) 5.88 T

Solution:

$$\begin{aligned}\frac{I}{C} &= \frac{1}{2} \epsilon_0 \cdot E_0^2 \\ \Rightarrow E_0 &= \sqrt{\frac{2I}{C\epsilon_0}} \\ \frac{E_0}{B_0} &= C \Rightarrow B_0 = \frac{E_0}{C} \\ \Rightarrow B_0 &= \sqrt{\frac{2I}{\epsilon_0 C^3}} = \sqrt{\frac{2 \times 0.092}{8.85 \times 10^{-12} \times 27 \times 10^{24}}} \\ &= 2.77 \times 10^{-8} \text{ T}\end{aligned}$$

Q17: The motion of a mass on a spring, with spring constant K is as shown in figure.



The equation of motion is given by $x(t) = A \sin \omega t + B \cos \omega t$ with $\omega = \sqrt{\frac{K}{m}}$

Suppose that at time $t = 0$, the position of mass is $x(0)$ and velocity $v(0)$, then its displacement can also be represented as $x(t) = C \cos(\omega t - \phi)$, where C and ϕ are :

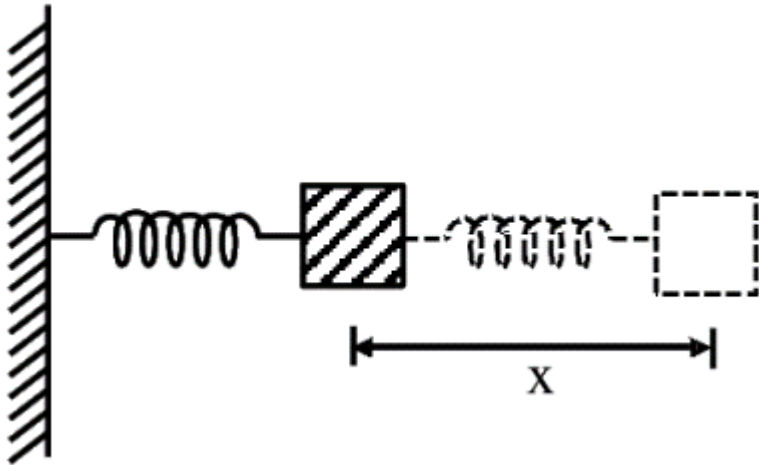
(A) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1} \left(\frac{v(0)}{x(0)\omega} \right)$

(B) $C = \sqrt{\frac{2v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1} \left(\frac{v(0)\omega}{2v(0)} \right)$

(C) $C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1} \left(\frac{x(0)\omega}{v(0)} \right)$

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$$(D) \quad C = \sqrt{\frac{v(0)^2}{\omega^2} + x(0)^2}, \phi = \tan^{-1} \left(\frac{v(0)}{x(0)\omega} \right)$$

Solution:

$$C \cos \phi = x(0)$$

$$tC\omega \sin \phi = v(0)$$

$$\left[\frac{v(0)}{\omega} \right]^2 + [x(0)]^2 = C^2$$

$$\tan \phi = \frac{v(0)}{x(0)\omega}$$

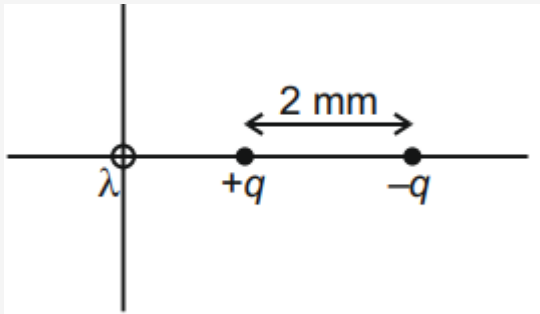
Q18: An electric dipole is placed on x – axis in proximity to a line charge of linear charge density $3.0 \times 10^{-6} \text{ C/m}$. Line charge is placed on z – axis and positive and negative charge of dipole is at a distance of 10 mm and 12 mm from the origin respectively. If total force of 4 N is exerted on the dipole, find out the amount of positive or negative charge of the dipole.

- (A) 815.1 nC
- (B) $8.8 \mu\text{C}$
- (C) 0.485, mC
- (D) $4.44 \mu\text{C}$

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- (C) 0.485, mC
- (D) $4.44 \mu\text{C}$

Solution:



$$|F| = q (E_1 - E_2)$$

$$= (q) 2k\lambda \left[\frac{2}{10 \times 12 \times 10^{-3}} \right]$$

$$4 = (q) \times 2 \times 9 \times 10^9 \times (3 \times 10^{-6}) \left[\frac{2}{120 \times 10^{-3}} \right]$$

$$\Rightarrow q = 4.44 \mu\text{C}$$

Q19: A body is projected vertically upwards from the surface of earth with a velocity sufficient enough to carry it to infinity. The time taken by it to reach height h is _____ S.

(A) $\sqrt{\frac{R_e}{2g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$

(B) $\sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$

(C) $\frac{1}{3} \sqrt{\frac{R_e}{2g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$

(D) $\frac{1}{3} \sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$

Q19: A body is projected vertically upwards from the surface of earth with a velocity sufficient enough to carry it to infinity. The time taken by it to reach height h is _____ S.

(A) $\sqrt{\frac{R_e}{2g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$

(B) $\sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$

(C) $\frac{1}{3} \sqrt{\frac{R_e}{2g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$

(D) $\frac{1}{3} \sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$

Solution:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = 0 \Rightarrow v = \sqrt{\frac{2GM}{r}}$$

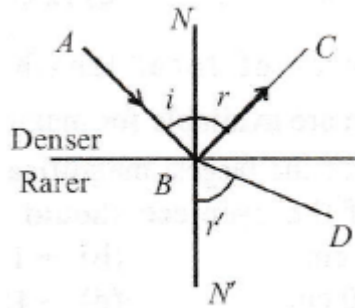
$$\frac{dr}{dt} = \sqrt{\frac{2GM}{r}}$$

$$\Rightarrow \int_{R_e}^{(R_e+h)} \sqrt{r} dr = \int_0^t \sqrt{2GM} dt$$

$$\Rightarrow \frac{2}{3} \left[(R_e + h)^{3/2} - R_e^{3/2} \right] = (t) \sqrt{2GM}$$

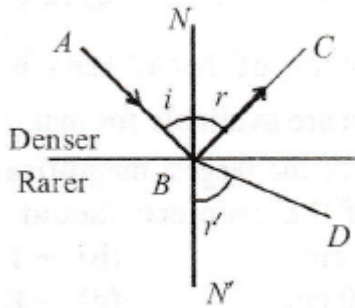
$$\Rightarrow t = \frac{1}{3} \sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

Q20: A ray of light from a denser medium strikes a rarer medium at an angle of incidence i (see figure). The reflected and refracted rays make an angle of 90° with each other. The angles of reflection and refraction are r and r' . The critical angle is



- (A) $\sin^{-1}(\tan r)$
- (B) $\sin^{-1}(\cot i)$
- (C) $\sin^{-1}(\tan r')$
- (D) $\tan^{-1}(\sin r)$

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- (A) $\sin^{-1}(\tan r)$
- (B) $\sin^{-1}(\cot i)$
- (C) $\sin^{-1}(\tan r')$
- (D) $\tan^{-1}(\sin r)$

Solution:

$$r + r' + 90^0 = 180^0$$

$$\therefore r' = 90^0 - r$$

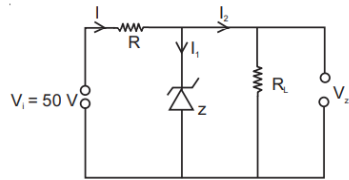
Further, $i = r$

Applying Snell's law, $\mu_D \sin i = \mu_R \sin r'$

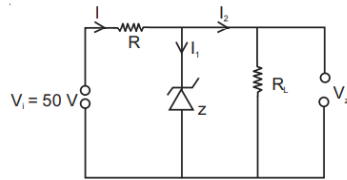
$$\text{or } \mu_D \sin r = \mu_R \sin(90^0 - r) = \mu_R \cos r$$

$$\therefore \frac{\mu_R}{\mu_D} = \tan r, \theta_C = \sin^{-1} \left(\frac{\mu_R}{\mu_D} \right) = \sin^{-1} (\tan r)$$

Q21: In a given circuit diagram, a 5 V Zener diode along with a series resistance is connected across a 50 V power supply. The minimum value of the resistance required, if the maximum Zener current is 90 mA will be _____ Ω

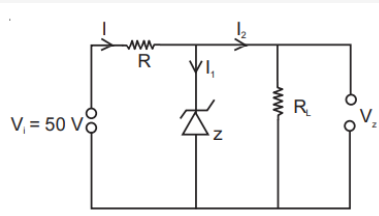


Q21: In a given circuit diagram, a 5 V Zener diode along with a series resistance is connected across a 50 V power supply. The minimum value of the resistance required, if the maximum Zener current is 90 mA will be _____ Ω



500.00

Solution:



$$I = \frac{50 - V_z}{R} = \frac{5}{R_L} + 90 \times 10^{-3}$$

For $R_L \rightarrow \infty$

$$R = 500\Omega$$

Q22: The position of the centre of mass of a uniform semi – circular wire of radius ' R ' placed in x – y plane with its centre at the origin and the line joining its ends as x – axis is given by $\left(0, \frac{xR}{\pi}\right)$.

Q22: The position of the centre of mass of a uniform semi – circular wire of radius ' R ' placed in x – y plane with its centre at the origin and the line joining its ends as x – axis is given by $\left(0, \frac{xR}{\pi}\right)$.

2.00

Solution:

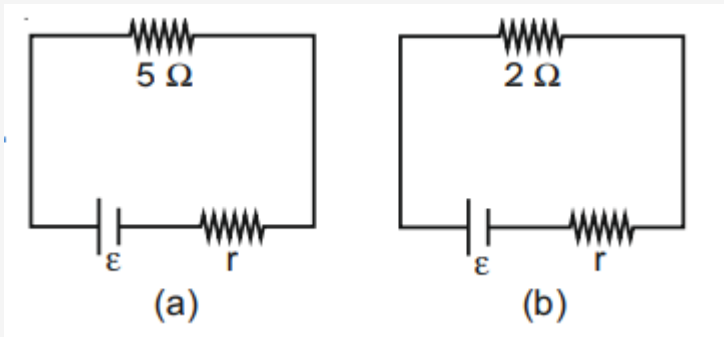
Centre of mass of half ring is located at a distance $\frac{2R}{\pi}$ from centre of the ring on its axis of symmetry so position of centre of mass in the given question will be $\left(0, \frac{2R}{\pi}\right) \Rightarrow |x| = 2$

Q23: In an electric circuit, a cell of certain emf provides a potential difference of 1.25 V across a load resistance of $5\ \Omega$. However, it provides a potential difference of 1 V across a load resistance of $2\ \Omega$. The emf of the cell is given by $\frac{x}{10}\text{ V}$. Then the value of x is _____.

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15.00

Solution:



In case (a), $\epsilon = \frac{1.25}{5}(5 + r)$

$\Rightarrow 4\epsilon = 5 + r$

In case (b), $\epsilon = \frac{1}{2}(2 + r)$

$\Rightarrow 2\epsilon = 2 + r$

From equation (1) & (2)

$2\epsilon = 3 \Rightarrow \epsilon = 1.5$

or $x = 15$

Q24: The total charge enclosed in an incremental volume of $2 \times 10^{-9} \text{ m}^3$ located at the origin is _____ nC, if electric flux density of its field is found as

$$\mathbf{D} = e^{-x} \sin y \hat{i} - e^{-x} \cos y \hat{j} + 2z \hat{k} \text{ C/m}^2.$$

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$$\mathbf{D} = e^{-x} \sin y \hat{i} - e^{-x} \cos y \hat{j} + 2z \hat{k} \text{ C/m}^2.$$

4.00

Solution:

$$\bar{D} = \epsilon_0 \bar{E}$$

$$\text{Div. } \bar{E} = \frac{\rho}{\epsilon_0}$$

$$\Rightarrow \text{div. } \bar{D} = \rho$$

$$\Rightarrow \frac{\partial}{\partial x}(e^{-x} \sin y) + \frac{\partial}{\partial y}(-e^{-x} \cos y) + \frac{\partial}{\partial z}(2z) = \rho$$

$$\Rightarrow \rho = 2 \text{ (a constant)}$$

$$V = 2 \times 10^{-9} \text{ m}^3$$

$$q = 2 \times 2 \times 10^{-9} = 4 \text{ nC}$$

Q25: Three particles P, Q and R are moving along the vectors $\vec{A} = \hat{i} + \hat{j}$, $\vec{B} = \hat{j} + \hat{k}$ and $\vec{C} = -\hat{i} + \hat{j}$ respectively. They strike on a point and start to move in different directions. Now particle P is moving normal to the plane which contains vector $\vec{A}$ and $\vec{B}$. Similarly particle Q is moving normal to the plane which contains vector $\vec{A}$ and $\vec{C}$. The angle between the direction of motion of P and Q is $\cos^{-1} \left(\frac{1}{\sqrt{x}} \right)$. Then the value of x is

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3.00

Solution:

$$\hat{n}_1 = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$$

$$\hat{n}_2 = \frac{\vec{A} \times \vec{C}}{|\vec{A} \times \vec{C}|} = \hat{k}$$

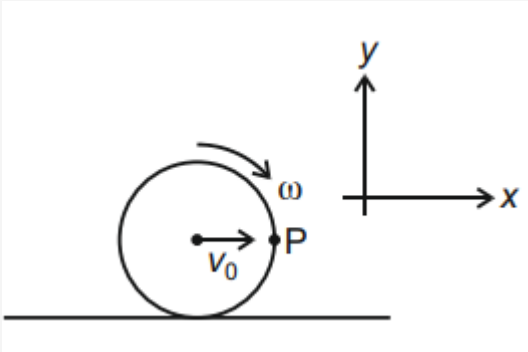
$$\cos \theta = \hat{n}_1 \cdot \hat{n}_2 = \frac{1}{\sqrt{3}}$$

Q26: The centre of a wheel rolling on a plane surface moves with a speed ν_0 . A particle on the rim of the wheel at the same level as the centre will be moving at a speed $\sqrt{x} \nu_0$. Then the value of x is _____.

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2.00

Solution:



$$|\omega| = \frac{v_0}{R}$$

$$\vec{V}_P = v_0 \hat{i} + \omega R (-\hat{j}) = v_0 \hat{i} - v_0 \hat{j}$$

$$|\vec{v}_P| = \sqrt{2} v_0$$

$$x = 02$$

Q27: A ray of light passing through a prism ($\mu = \sqrt{3}$) suffers minimum deviation. It is found that the angle of incidence is double the angle of refraction within the prism. Then, the angle of prism is _____ (in degrees)

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60.00

Solution:

For minimum deviation $r_1 = r_2 = A/2$

Given $i = 2r$

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin 2r}{\sin r}$$

$$\Rightarrow \cos r = \frac{\mu}{2}$$

$$\Rightarrow r = 30^\circ$$

$$\Rightarrow A = 60^\circ$$

Q28: The area of cross – section of a railway track is 0.01 m^2 . The temperature variation is 10°C . Coefficient of linear expansion of material of track is $10^{-5} / ^\circ \text{C}$. The energy stored per meter in the track is _____ J/m.

(Young's modulus of material of track is 10^{11} Nm^{-2})

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(Young's modulus of material of track is 10^{11} Nm^{-2})

5.00

Solution:

As the tracks won't be allowed to expand linearly, the rise in temperature would lead to developing thermal stress in track.

$$\frac{(\text{Stress})}{Y} = \alpha \Delta T \text{ or } \sigma = Y \alpha \Delta T$$

$$\text{Energy stored per unit volume} = \frac{1}{2} \frac{\sigma^2}{Y}$$

$$\Rightarrow \text{Energy stored per unit length} = \frac{A \sigma^2}{2Y}$$

$$= \frac{A}{2} \times Y \alpha^2 \Delta T^2$$

$$= \frac{10^{-2} \times 10^{11} \times 10^{-10} \times 100}{2} = 5 \text{ J/m}$$

Q29: Three students S_1 , S_2 and S_3 perform an experiment for determining the acceleration due to gravity (g) using a simple pendulum. They use different lengths of pendulum and record time for different numbers of oscillations. The observations are as shown in the table.

Student No.	Length of pendulum (cm)	No. of oscillations (n)	Total time for n oscillations	Time period (s)
1.	64.0	8	128.0	16.0
2.	64.0	4	64.0	16.0
3.	20.0	4	36.0	9.0

(Least count of length = 0.1 m least count for time = 0.1 s)

If E_1 , E_2 and E_3 are the percentage errors in ' g ' for students 1, 2 and 3 respectively, then the minimum percentage error is obtained by students no._____.

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If E_1 , E_2 and E_3 are the percentage errors in ' g ' for students 1, 2 and 3 respectively, then the minimum percentage error is obtained by students no._____.

Solution:

$$T = \frac{t}{n} = 2\pi\sqrt{\frac{l}{g}}$$

$$\Rightarrow g = \frac{4\pi^2 l}{T^2}$$

$$\Rightarrow \frac{\Delta g}{g} \times 100 = \frac{\Delta l}{l} \times 100 + 2 \frac{\Delta T}{T} \times 100$$

$$= \left(\frac{\Delta l}{l} + \frac{2\Delta t}{nT} \right) 100\%$$

$$E_1 = \frac{20}{64}\%$$

$$E_2 = \frac{30}{64}\%$$

$$E_3 = \frac{19}{18}\%$$

Q30: In 5 minutes, a body cools from 75°C to 65°C at room temperature of 25°C . The temperature of body at the end of next 5 minutes is _____ $^{\circ}\text{C}$.

Q30: In 5 minutes, a body cools from $75^{\circ}C$ to $65^{\circ}C$ at room temperature of $25^{\circ}C$. The temperature of body at the end of next 5 minutes is _____ $^{\circ}C$.

57.00

Solution:

$$\frac{75-65}{5} = k \left(\frac{75+65}{2} - 25 \right)$$

$$\Rightarrow k = \frac{2}{45}$$

$$\frac{65-T}{5} = k \left(\frac{65+T}{2} - 25 \right)$$

$$\Rightarrow T = 57^{\circ}C$$

Q31: The water having more dissolved O_2 is:

- (A) boiling water
- (B) water at $80^\circ C$
- (C) polluted water
- (D) water at $4^\circ C$

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Solution:

As the temperature increases, the solubility of gases in the liquid decreases. Thus, water at $4^\circ C$ will have maximum dissolved oxygen.

Polluted water has less oxygen concentration than normal water as a large population of bacteria decomposes organic matter present in water. They consume oxygen dissolved in water.

Q32: Which one of the following statements for D.I. Mendeleev, is **incorrect**?

- (A) He authored the textbook – Principles of Chemistry.
- (B) At the time, he proposed Periodic Table of elements structure of atom was known.
- (C) Element with atomic number 101 is named after him.
- (D) He invented accurate barometer.

Q32: Which one of the following statements for D.I. Mendeleev, is incorrect?

- (A) He authored the textbook – Principles of Chemistry.
- (B) At the time, he proposed Periodic Table of elements structure of atom was known.**
- (C) Element with atomic number 101 is named after him.
- (D) He invented accurate barometer.

Solution:

- (a) Dmitri Ivanovich Mendeleev taught at the University of St. Petersburg where he was appointed Professor of General Chemistry in 1867. Preliminary work for his great textbook “Principles of Chemistry” led Mendeleev to propose the Periodic Law and to construct his Periodic Table of elements.
- (b) At that time, the structure of atom was unknown and Mendeleev’s idea to consider that the properties of the elements were in some way related to their atomic masses was a very imaginative one.
- (c) Mendeleev’s name has been immortalized by naming the element with atomic number 101, as Mendelevium.
- (d) Mendeleev was a versatile genius. He worked on many problems connected with Russia’s natural resources. He invented an accurate barometer.

Q33:

Which purification technique is used for the organic compound that decomposes near its boiling point?

- (A) Simple distillation
- (B) Steam distillation
- (C) Fractional distillation
- (D) Reduced pressure distillation

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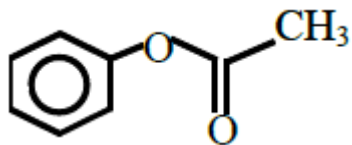
- (A) Simple distillation
- (B) Steam distillation
- (C) Fractional distillation
- ☒ (D) Reduced pressure distillation

Solution:

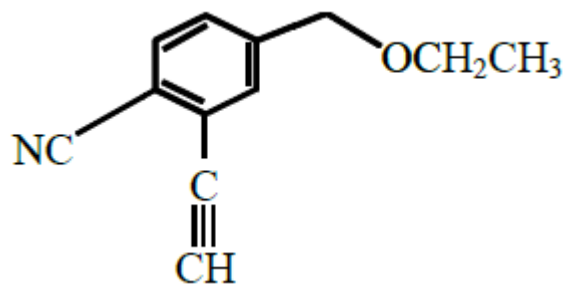
Reduced pressure distillation or vacuum distillation is used for the purification of organic liquids which decompose at or below their boiling point.

Q34: Which of the following compounds will provide a tertiary alcohol on reaction with excess of CH_3MgBr followed by hydrolysis?

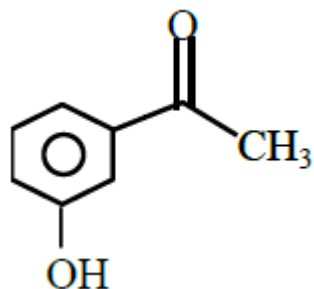
(A)



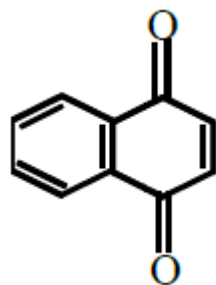
(B)



(C)

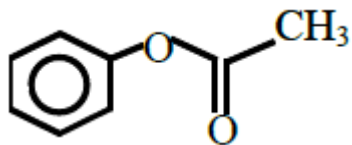


(D)

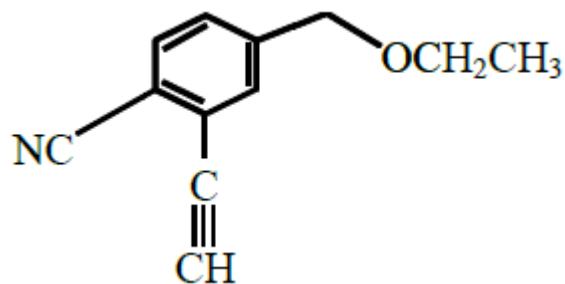


Q34: Which of the following compounds will provide a tertiary alcohol on reaction with excess of CH_3MgBr followed by hydrolysis?

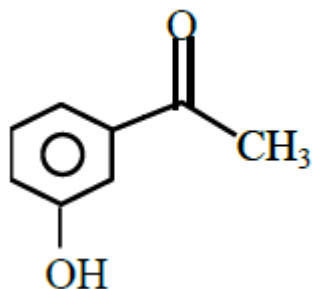
(A)



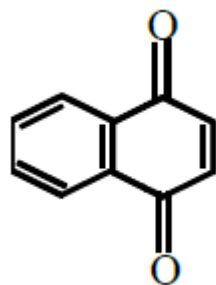
(B)



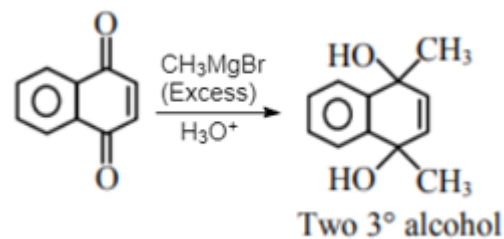
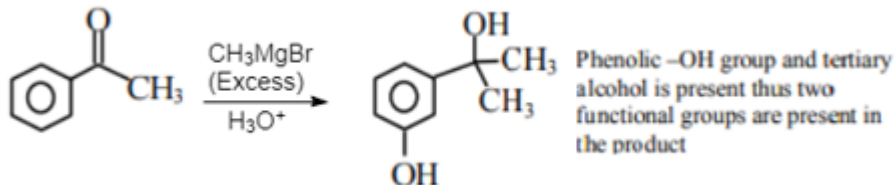
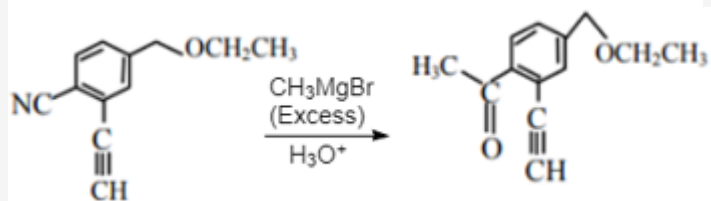
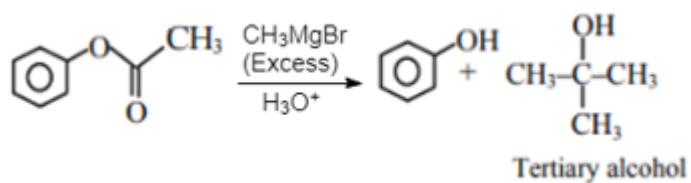
(C)



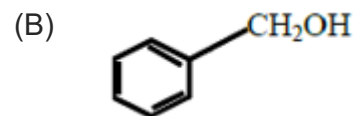
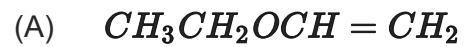
(D)



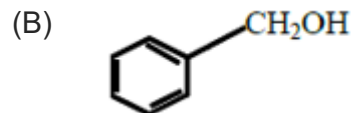
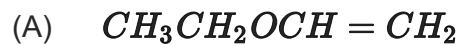
Solution:



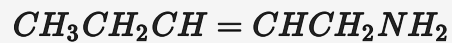
Q35: Which of the following compounds does not exhibit resonance?



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Solution:



No conjugation is present.

Q36: Match List-I with List-II

List-I (Elements)	List-II (Properties)
(a) Ba	(i) Organic solvent soluble compounds
(b) Ca	(ii) Outer electronic configuration $6s^2$
(c) Li	(iii) Oxalate insoluble in water
(d) Na	(iv) Formation of very strong monoacidic base

Choose the **correct** answer from the options given below:

- (A) (a)-(ii), (b)-(iii), (c)-(i) and (d)-(iv)
- (B) (a)-(iv), (b)-(i), (c)-(ii) and (d)-(iii)
- (C) (a)-(iii), (b)-(ii), (c)-(iv) and (d)-(i)
- (D) (a)-(i), (b)-(iv), (c)-(ii) and (d)-(iii)

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- (B) (a)-(iv), (b)-(i), (c)-(ii) and (d)-(iii)
- (C) (a)-(iii), (b)-(ii), (c)-(iv) and (d)-(i)
- (D) (a)-(i), (b)-(iv), (c)-(ii) and (d)-(iii)

Solution:

Option (a) has all correct matchings because -

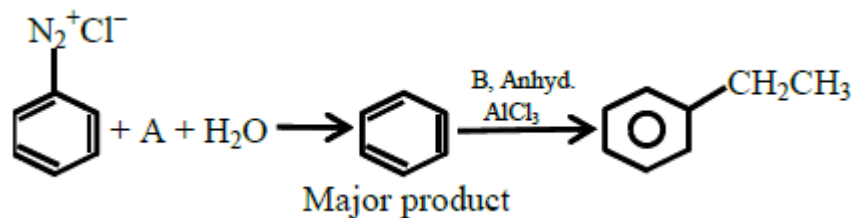
(a) Ba has electronic configuration $[\text{Xe}] 6s^2$.

(b) Ca forms insoluble oxalate.

(c) Li forms mostly covalent compounds which are soluble in organic solvents.

(d) Na forms a base NaOH which is monoacidic.

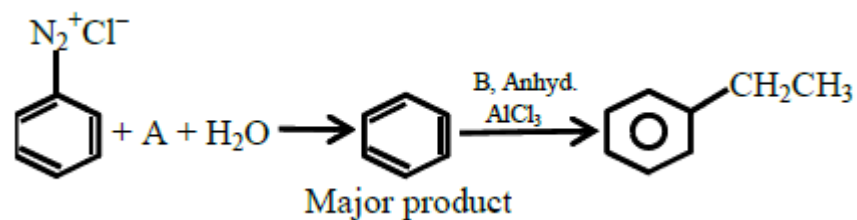
Q37:



In the chemical reactions given above A and B respectively are:

- (A) H_3PO_2 and $\text{CH}_3\text{CH}_2\text{Cl}$
- (B) $\text{CH}_3\text{CH}_2\text{OH}$ and H_3PO_2
- (C) H_3PO_2 and $\text{CH}_3\text{CH}_2\text{OH}$
- (D) $\text{CH}_3\text{CH}_2\text{Cl}$ and H_3PO_2

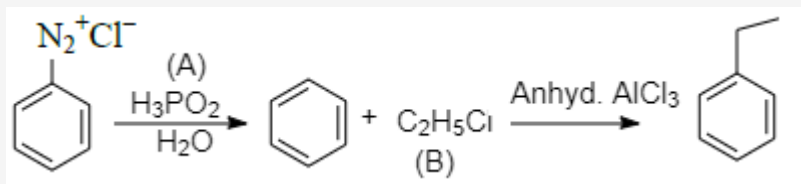
Q37:



In the chemical reactions given above A and B respectively are:

- (A) H_3PO_2 and CH_3CH_2Cl
- (B) CH_3CH_2OH and H_3PO_2
- (C) H_3PO_2 and CH_3CH_2OH
- (D) CH_3CH_2Cl and H_3PO_2

Solution:



Q38: Isotope(s) of hydrogen which emits low energy β^- particles with $t_{1/2}$ value > 12 years is/are

- (A) Protium
- (B) Tritium
- (C) Deuterium
- (D) Deuterium and Tritium

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- (A) Protium
- ☒ (B) Tritium
- (C) Deuterium
- (D) Deuterium and Tritium

Solution:

Of the three isotopes of hydrogen, only tritium is radioactive and emits low energy β^- particles ($t_{1/2} = 12.33$ years)

Q39: Match **List-I** and **List-II** :

List-I (Species)	List-II (Hybrid Orbitals)
(a) SF_4	(i) sp^3d^2
(b) IF_5	(ii) d^2sp^3
(c) NO_2^+	(iii) sp^3d
(d) NH_4^+	(iv) sp^3
	(v) sp

Choose the **correct** answer from the options given below:

- (A) (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)
- (B) (a)-(ii), (b)-(i), (c)-(iv) and (d)-(v)
- (C) (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv)
- (D) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v)

Q39: Match List-I and List-II :

List-I (Species)	List-II (Hybrid Orbitals)
(a) SF_4	(i) sp^3d^2
(b) IF_5	(ii) d^2sp^3
(c) NO_2^+	(iii) sp^3d
(d) NH_4^+	(iv) sp^3
	(v) sp

Choose the **correct** answer from the options given below:

- (A) (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)
- (B) (a)-(ii), (b)-(i), (c)-(iv) and (d)-(v)
- (C)** (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv)
- (D) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v)

Solution:

Species	Number of hybrid orbitals	Hybridisation
SF_4	$\frac{1}{2}[6 + 4] = 5$	sp^3d
IF_5	$\frac{1}{2}[7 + 5] = 6$	sp^3d^2
NO_2^+	$\frac{1}{2}[5 - 1] = 2$	sp
NH_4^+	$\frac{1}{2}[5 + 4 - 1] = 4$	sp^3

Q40: When silver nitrate solution is added to potassium iodide solution then the sol produced is:

- (A) AgI/I^-
- (B) AgI/Ag^+
- (C) KI/NO_3^-
- (D) $AgNO_3/NO_3^-$

Q40: When silver nitrate solution is added to potassium iodide solution then the sol produced is:

- (A) AgI/I^-
- (B) AgI/Ag^+
- (C) KI/NO_3^-
- (D) $AgNO_3/NO_3^-$

Solution:

When silver nitrate is added to potassium iodide solution, all of silver nitrate is converted into AgI and it adsorbs the common ion I^- from the solution to form the negatively charged colloidal particles AgI / I^-

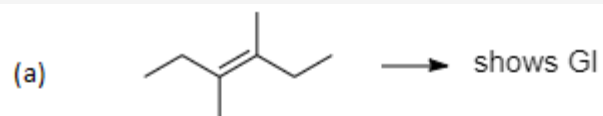
Q41: Which of the following molecules does not show stereoisomerism?

- (A) 3, 4-Dimethylhex-3-ene
- (B) 3-Methylhex-1-ene
- (C) 3-Ethylhex-3-ene
- (D) 4-Methylhex-1-ene

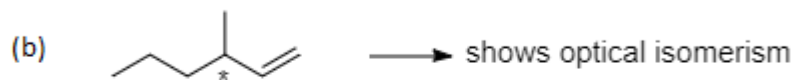
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- (C) 3-Ethylhex-3-ene
- (D) 4-Methylhex-1-ene

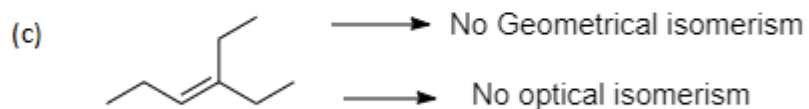
Solution:



3, 4-Dimethylhex-3-ene



3-Methylhex-1-ene



3-Ethylhex-3-ene



4-Methylhex-1-ene

Q42: Given below are the statements about diborane -

(a) Diborane is prepared by the oxidation of NaBH_4 with I_2

(b) Each boron atom is in sp^2 hybridized state

(c) Diborane has one bridged 3 centre-2-electron bond

(d) Diborane is a planar molecule

The option with **correct** statement(s) is -

(A) (c) and (d) only

(B) (a) only

(C) (c) only

(D) (a) and (b) only

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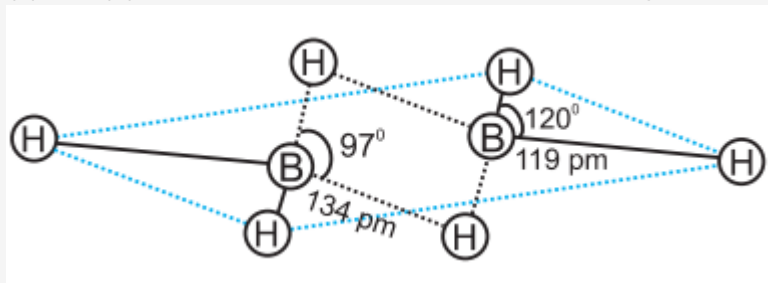
(D) (a) and (b) only

Solution:

(a) A convenient laboratory method for the preparation of diborane involves the oxidation of sodium borohydride with iodine –
 $2\text{NaBH}_4 + \text{I}_2 \rightarrow \text{B}_2\text{H}_6 + 2\text{NaI} + \text{H}_2$

(b) Each B atom uses sp^3 hybrids for bonding. Out of the four sp^3 hybrids on each B atom, one is without an electron.

(c) and (d): The structure of diborane is shown in Fig.



The four terminal hydrogen atoms and the two boron atoms lie in one plane. Above and below this plane, there are two bridging hydrogen atoms. The four terminal B-H bonds are regular two centre-two electron bonds while the two bridge (B-H-B) bonds are different and can be described in terms of three centre-two electron bonds.

Q43: Which one of the following group-15 hydrides is the strongest reducing agent?



Q43: Which one of the following group-15 hydrides is the strongest reducing agent?

(A) AsH_3

(B) BiH_3

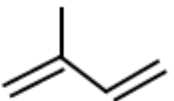
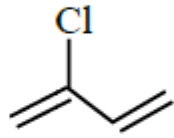
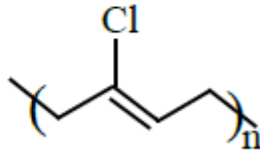
(C) PH_3

(D) SbH_3

Solution:

The stability of group-15 hydrides decreases from NH_3 to BiH_3 which can be observed from their bond dissociation enthalpy. Consequently, the reducing character of the hydrides increases from NH_3 to BiH_3 .

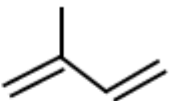
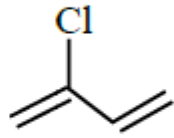
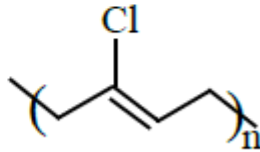
Q44: Match **List-I** with **List-II**:

List-I	List-II
(a) Chloroprene	(i) 
(b) Neoprene	(ii) 
(c) Acrylonitrile	(iii) 
(d) Isoprene	(iv) $\text{CH}_2 = \text{CH} - \text{CN}$

Choose the **correct** answer from the options given below:

- (A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (B) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q44: Match **List-I** with **List-II**:

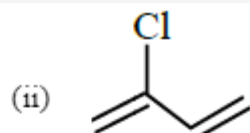
List-I	List-II
(a) Chloroprene	(i) 
(b) Neoprene	(ii) 
(c) Acrylonitrile	(iii) 
(d) Isoprene	(iv) $\text{CH}_2 = \text{CH} - \text{CN}$

Choose the **correct** answer from the options given below:

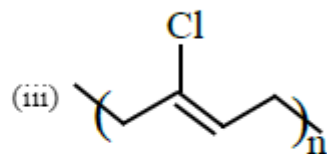
- (A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (B) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
- (D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Solution:

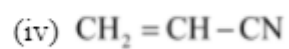
(a) Chloroprene



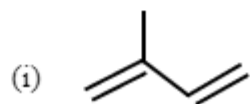
(b) Neoprene



(c) Acrylonitrile



(d) Isoprene



Q45: The set having ions which are coloured and paramagnetic both is -

- (A) Cu^{2+} , Cr^{3+} , Sc^{+}
- (B) Cu^{2+} , Zn^{2+} , Mn^{4+}
- (C) Sc^{3+} , V^{5+} , Ti^{4+}
- (D) Ni^{2+} , Mn^{7+} , Hg^{2+}

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- (D) Ni^{2+} , Mn^{7+} , Hg^{2+}

Solution:

The option (a) has all ions which are coloured and paramagnetic as well due to presence of unpaired electrons.

Cu^{2+} : $[\text{Ar}] 3d^9$

Cr^{3+} : $[\text{Ar}] 3d^3$

Sc^{+} : $[\text{Ar}] 3d^1 4s^1$

Q46: Thiamine and pyridoxine are also known respectively as:

- (A) Vitamin ***B*₂** and Vitamin E
- (B) Vitamin E and Vitamin ***B*₂**
- (C) Vitamin ***B*₆** and Vitamin ***B*₂**
- (D) Vitamin ***B*₁** and Vitamin ***B*₆**

Q46: Thiamine and pyridoxine are also known respectively as:

- (A) Vitamin B_2 and Vitamin E
- (B) Vitamin E and Vitamin B_2
- (C) Vitamin B_6 and Vitamin B_2
- (D) Vitamin B_1 and Vitamin B_6

Solution:

Vitamin B_1 is also known as Thiamine while vitamin B_6 is known as Pyridoxine.

Q47: Sulphide ion is soft base, and its ores are common for metals.

(a) Pb, (b) Al, (c) Ag, (d) Mg

Choose the **correct** answer from the options given below:

(A) (a) and (c) only

(B) (a) and (d) only

(C) (a) and (b) only

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Choose the **correct** answer from the options given below:

(A) (a) and (c) only

(B) (a) and (d) only

(C) (a) and (b) only

(D) (c) and (d) only

Solution:

(a) Common ore for Pb is galena, PbS .

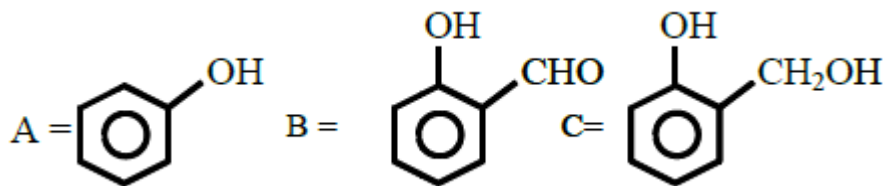
(b) Common ore for Al is bauxite, $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$.

(c) Common ore for Ag is Argentite or silver glance, Ag_2S .

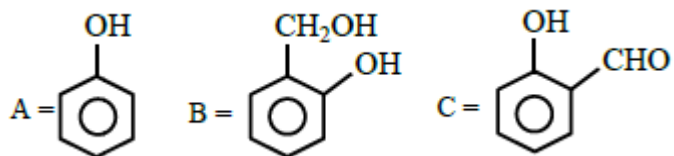
(d) Common ore for Mg is Carnalite, $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$.

Q48: An organic compound A (C_6H_6O) gives dark green coloration with ferric chloride. On treatment with $CHCl_3$ and KOH, followed by acidification gives compound B. Compound B can also be obtained from compound C on reaction with pyridinium chlorochromate (PCC). Identify A, B and C.

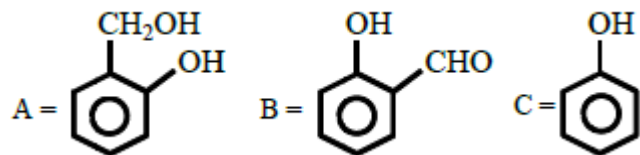
(A)



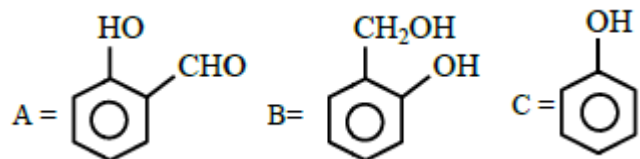
(B)



(C)

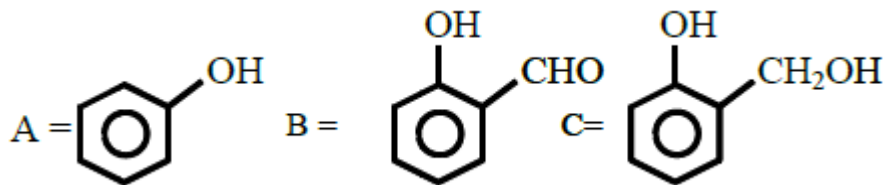


(D)

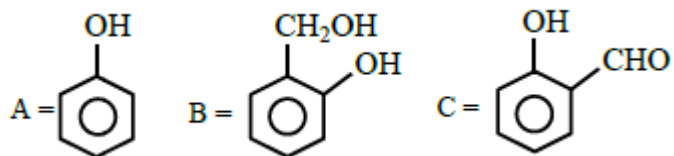


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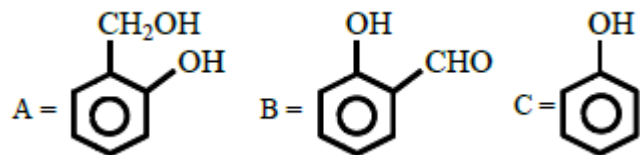
(A)



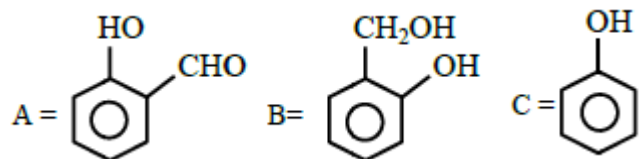
(B)



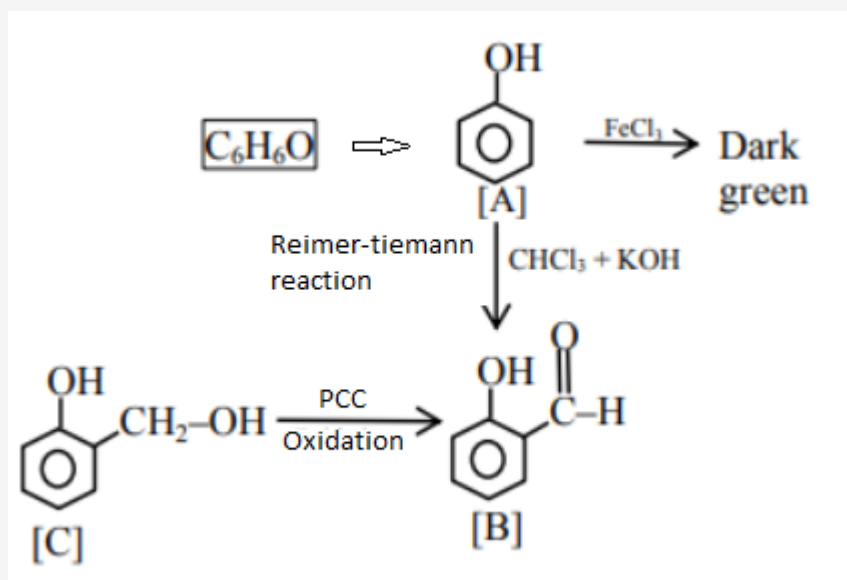
(C)



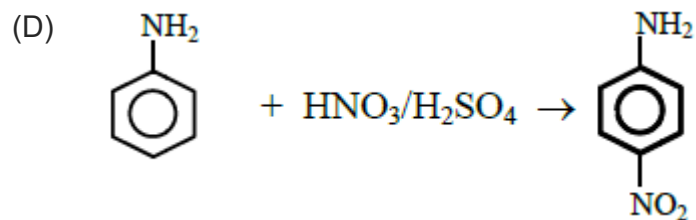
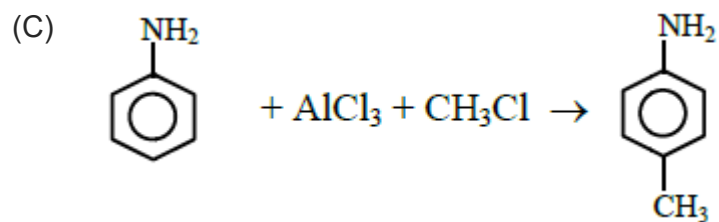
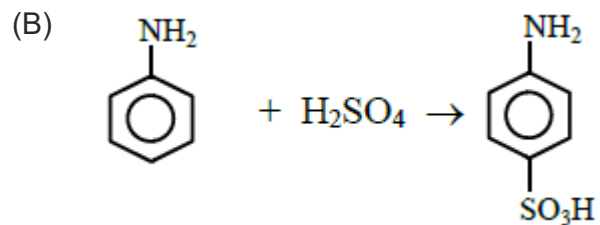
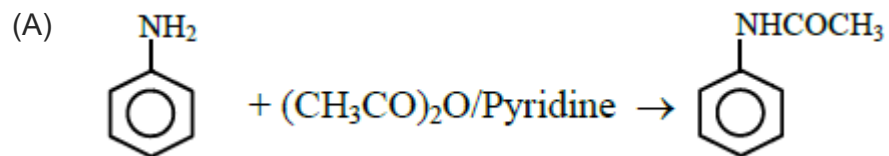
(D)



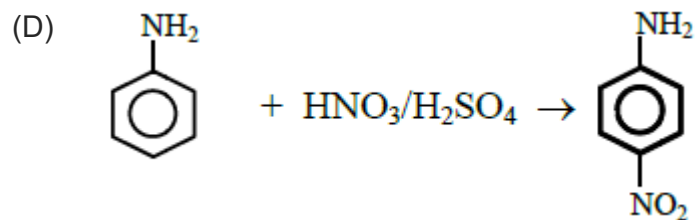
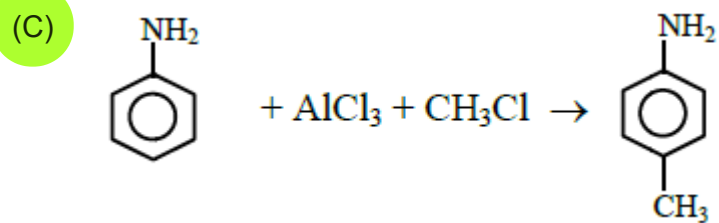
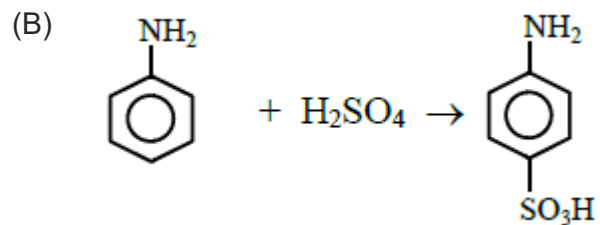
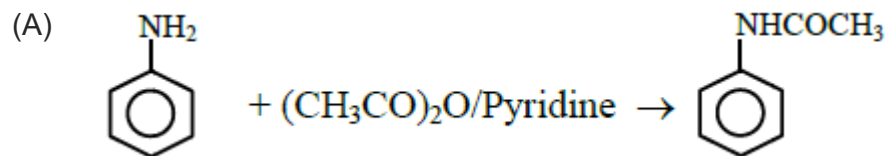
Solution:



Q49: Which one of the following reactions does not occur?



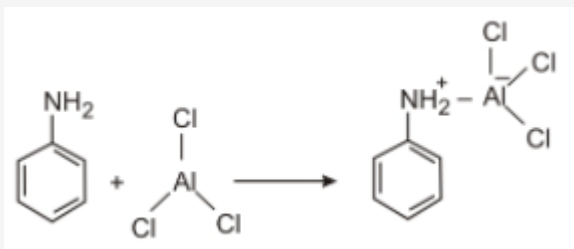
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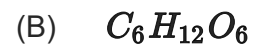
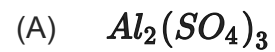
Solution:

Aniline does not undergo Friedel craft reaction.

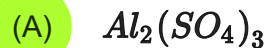
Due to the lone pair of electrons on N, aniline is a strong lewis base. Also, the catalyst used (AlCl_3) is a very strong lewis acid.



Q50: Which one of the following 0.06 M aqueous solutions has lowest freezing point?



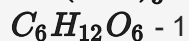
Q50: Which one of the following 0.06 M aqueous solutions has lowest freezing point?



Solution:

As the value of ΔT_f increases, the freezing point of the solution decreases. Since all the given solutions have the same concentration, the one with maximum value of van't Hoff factor for the solute will have minimum freezing point. The values are

–



Thus, among given solutes, aq. solution of $Al_2(SO_4)_3$ will have lowest freezing point.

Q51: The total number of unpaired electrons present in $[Co(NH_3)_6]Cl_2$ and $[Co(NH_3)_6]Cl_3$ is

Q51: The total number of unpaired electrons present in $[Co(NH_3)_6]Cl_2$ and $[Co(NH_3)_6]Cl_3$ is

1.00

Solution:

$[Co(NH_3)_6]Cl_2$: $Co^{2+} = [Ar] 3d^7$. In presence of strong field ammine ligand, the electrons will be paired and one unpaired electron will be excited to the 4d orbitals so that the hybridization is d^2sp^3 and thus the number of unpaired electrons = 1.

$[Co(NH_3)_6]Cl_3$: $Co^{3+} = [Ar] 3d^6$. In presence of strong field ammine ligand, all the electrons will be paired and thus the number of unpaired electrons = 0.

Q52: Methylation of 10 g of benzene gave 9.2 g of toluene. Calculate the percentage yield of toluene _____. (Nearest integer)

Q52: Methylation of 10 g of benzene gave 9.2 g of toluene. Calculate the percentage yield of toluene _____. (Nearest integer)

78.00

Solution:

One mole of benzene will form one mole of toluene at 100% yield.

i.e., 78 g benzene will produce 92 g of toluene at 100% yield.

or 10 g benzene will produce $\frac{92}{78} \times 10$ g of toluene at 100% yield.

Thus, percentage yield = $\frac{\text{actual amount formed}}{\text{theoretical amount formed}} \times 100$

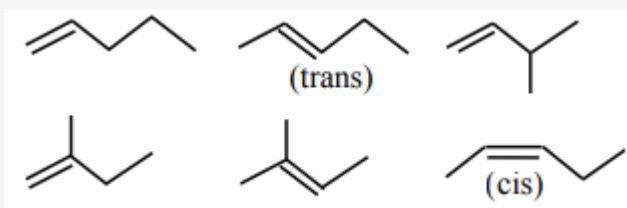
Therefore, %yield = $\frac{9.2}{\frac{92}{78} \times 10} \times 100 = 78\%$

Q53: The number of acyclic structural isomers (including geometrical isomers) for pentene are

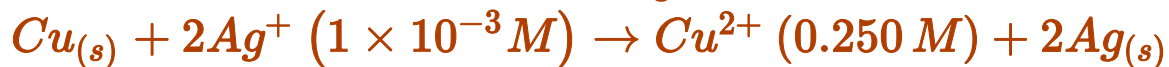
Q53: The number of acyclic structural isomers (including geometrical isomers) for pentene are

6.00

Solution:



Q54: Assume a cell with the following reaction



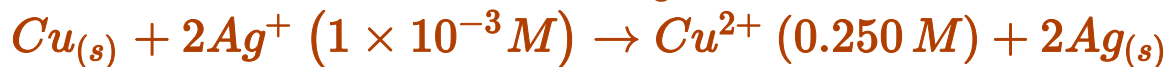
$$E_{cell}^{\ominus} = 2.97 V$$

E_{cell} for the above reaction is _____ V.

(Nearest integer)

[Given: $\log 2.5 = 0.3979$, $T = 298 K$]

Q54: Assume a cell with the following reaction



$$E_{cell}^{\ominus} = 2.97 V$$

E_{cell} for the above reaction is _____ V.

(Nearest integer)

[Given: $\log 2.5 = 0.3979$, $T = 298 K$]

3.00

Solution:

Using Nernst equation –

$$E_{cell} = E_{cell}^{\circ} - \frac{0.06}{2} \log \frac{[Cu^{2+}]}{[Ag^+]^2}$$

$$\Rightarrow E_{cell} = 2.97 - 0.03 \times \log \frac{0.25}{10^{-6}}$$

$$\Rightarrow E_{cell} = 2.97 - 0.03 \times \log(2.5 \times 10^5)$$

$$\Rightarrow E_{cell} = 2.97 - 0.03 \times (5 + 0.3979)$$

$$\Rightarrow E_{cell} = 2.97 - 0.162$$

$$\Rightarrow E_{cell} = 2.808 V$$

Q55: Value of K_P for the equilibrium reaction $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ at 288 K is 47.9. The K_C for this reaction at same temperature is _____. (Nearest integer)
($R = 0.083 \text{ L bar } K^{-1} \text{ mol}^{-1}$)

Q55: Value of K_P for the equilibrium reaction $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ at 288 K is 47.9. The K_C for this reaction at same temperature is _____. (Nearest integer)
($R = 0.083 \text{ L bar } K^{-1} \text{ mol}^{-1}$)

2.00

Solution:

We know that $K_p = K_c(RT)^{\Delta n_g}$

$$\Rightarrow 47.9 = K_c(0.083 \times 288)^1$$

$$\Rightarrow 47.9 = K_c \times 23.904$$

$$\Rightarrow K_c = \frac{47.9}{23.904} \approx 2$$

Q56: If the standard molar enthalpy change for combustion of graphite powder is $-2.48 \times 10^2 \text{ kJ mol}^{-1}$, the amount of heat generated on combustion of 1 g of graphite powder is _____ kJ. (Nearest integer)

Q56: If the standard molar enthalpy change for combustion of graphite powder is $-2.48 \times 10^2 \text{ kJ mol}^{-1}$, the amount of heat generated on combustion of 1 g of graphite powder is _____ kJ. (Nearest integer)

21.00

Solution:

$\Delta_c H^\circ$ (standard molar enthalpy change for combustion) is defined as the amount of heat liberated on combustion of one mole of a substance.

12 g of graphite powder on combustion gives 2.48×10^2 kJ energy

Thus 1 g of graphite powder on combustion will give $\frac{2.48 \times 100}{12} = 20.67$ kJ energy.

Q57: A copper complex crystallizing in a CCP lattice with a cell edge of 0.4518 nm has been revealed by employing X-ray diffraction studies. The density of a copper complex is found to be 7.62 g cm^{-3} . The molar mass of copper complex is _____ g mol^{-1} . (Nearest integer)
[Given: $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$]

Q57: A copper complex crystallizing in a CCP lattice with a cell edge of 0.4518 nm has been revealed by employing X-ray diffraction studies. The density of a copper complex is found to be 7.62 g cm^{-3} . The molar mass of copper complex is _____ g mol^{-1} . (Nearest integer)
[Given: $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$]

106.00

Solution:

We know that $d = \frac{ZM}{N_A a^3}$

$$\Rightarrow 7.62 = \frac{4M}{6.022 \times 10^{23} \times (0.4518 \times 10^{-7})^3}$$

$$\Rightarrow 7.62 = \frac{4M}{0.5554 \times 10^{23} \times 10^{-21}}$$

$$\Rightarrow M = \frac{7.62 \times 0.5554 \times 10^2}{4} = 105.8 \text{ g mol}^{-1}$$

Q58: Number of electrons that Vanadium ($Z = 23$) has in p-orbitals is equal to _____

Q58: Number of electrons that Vanadium ($Z = 23$) has in p-orbitals is equal to _____

12.00

Solution:

The electronic configuration of Vanadium is $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^3$.

Thus, the number of electrons in p-orbitals = 12.



In the above first order reaction the initial concentration of N_2O_5 is $2.40 \times 10^{-2} \text{ mol L}^{-1}$ at 318 K. The concentration of N_2O_5 after 1 hour was $1.60 \times 10^{-2} \text{ mol L}^{-1}$. The rate constant of the reaction at 318 K is _____ $\times 10^{-3} \text{ min}^{-1}$. (Nearest integer)

[Given: $\log 3 = 0.477$, $\log 2 = 0.3010$]



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[Given: $\log 3 = 0.477$, $\log 2 = 0.3010$]

7.00

Solution:

Using the integrated rate law for first order reaction –

$$\log \frac{[R]_0}{[R]_t} = \frac{kt}{2.303}$$

$$\Rightarrow \log \frac{2.4 \times 10^{-2}}{1.6 \times 10^{-2}} = \frac{k \times 60}{2.303}$$

$$\Rightarrow \log \frac{3}{2} = \frac{k \times 60}{2.303}$$

$$\Rightarrow (0.4771 - 0.3010) = \frac{k \times 60}{2.303}$$

$$\Rightarrow k = \frac{0.1761 \times 2.303}{60} = 0.00676 \text{ min}^{-1} \text{ or } 6.76 \times 10^{-3} \text{ min}^{-1}$$

Q60: If the concentration of glucose ($C_6H_{12}O_6$) in blood is 0.72 g L^{-1} , the molarity of glucose in blood is _____ $\times 10^{-3} \text{ M}$. (Nearest integer)

[Given: Atomic mass of C = 12, H = 1, O = 16 u]

Q60: If the concentration of glucose ($C_6H_{12}O_6$) in blood is 0.72 g L^{-1} , the molarity of glucose in blood is _____ $\times 10^{-3} \text{ M}$. (Nearest integer)

[Given: Atomic mass of C = 12, H = 1, O = 16 u]

4.00

Solution:

The mass of glucose present in 1 L of blood = 0.72 g

Thus, moles of glucose present in 1 L of blood = $\frac{0.72}{180} = 0.004 = 4 \times 10^{-3}$

Since molarity is defined as the moles of solute present in 1 L solution, molarity of the given solution = $4 \times 10^{-3} \text{ M}$

Q61: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} -\frac{4}{3}x^3 + 2x^2 + 3x, & x > 0 \\ 3xe^x, & x \leq 0 \end{cases}$$

Then f is increasing function in the interval.

(A) $\left(-1, \frac{3}{2}\right)$

(B) $\left(\frac{-1}{2}, 2\right)$

(C) $(0, 2)$

(D) $(-3, -1)$

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(C) $(0, 2)$

(D) $(-3, -1)$

Solution:

For $x > 0$, $f'(x) = -4x^2 + 4x + 3$

$f(x)$ is increasing in $\left(-\frac{1}{2}, \frac{3}{2}\right)$

$\Rightarrow f(x)$ is increasing in $\left(0, \frac{3}{2}\right)$ ($\because x > 0$)

For $x \leq 0$, $f'(x) = 3e^x(1+x)$

$f'(x) > 0 \quad \forall x \in (-1, 0)$

$\Rightarrow f(x)$ is increasing in $(-1, 0)$

So, in complete domain $f(x)$ is increasing in $\left(-1, \frac{3}{2}\right)$

Q62: Let $[x]$ denote the greatest integer less than or equal to x . Then, the values of $x \in \mathbf{R}$ satisfying the equation $[e^x]^2 + [e^x + 1] - 3 = 0$ lie in the interval:

- (A) $[\ln 2, \ln 3)$
- (B) $[0, \frac{1}{e})$
- (C) $[0, \ln 2)$
- (D) $[1, e)$

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(A) $[\ln 2, \ln 3)$

(B) $[0, \frac{1}{e})$

(C) $[0, \ln 2)$

(D) $[1, e)$

Solution:

$$[e^x]^2 + [e^x + 1] - 3 = 0$$

$$\Rightarrow [e^x]^2 + [e^x] + 1 - 3 = 0$$

$$\text{Let } [e^x] = t$$

$$\Rightarrow t^2 + t - 2 = 0$$

$$\Rightarrow t = -2, 1$$

$$[e^x] = -2 \text{ (Not possible)}$$

$$\text{or } [e^x] = 1 \quad \therefore 1 \leq e^x < 2$$

$$\Rightarrow \ln(1) \leq x < \ln(2)$$

$$\Rightarrow 0 \leq x < \ln(2)$$

$$\Rightarrow x \in [0, \ln 2)$$

Q63: Let n denote the number of solutions of the equation $z^2 + 3\bar{z} = 0$, z is a complex number. Then the value of $\sum_{k=0}^{\infty} \frac{1}{n^k}$ is equal to:

(A) 1

(B) 2

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

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(A) 1

(B) 2

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

Solution:

$$z^2 + 3\bar{z} = 0$$

Put $z = x + iy$

$$\Rightarrow x^2 - y^2 + 2ixy + 3(x - iy) = 0$$

$$\Rightarrow (x^2 - y^2 + 3x) + i(2xy - 3y) = 0 + i0$$

$$\therefore x^2 - y^2 + 3x = 0 \quad \dots (1)$$

$$2xy - 3y = 0 \dots$$

$$x = \frac{3}{2}, y = 0$$

Put $x = \frac{3}{2}$ in equation (1)

$$\frac{9}{4} - y^2 + \frac{9}{2} = 0$$

$$y^2 = \frac{27}{4} \Rightarrow y = \pm \frac{3\sqrt{3}}{2}$$

$$\therefore (x, y) = \left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right), \left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$$

Put $y = 0 \Rightarrow x^2 - 0 + 3x = 0$

$$x = 0, -3$$

$$\therefore (x, y) = (0, 0), (-3, 0)$$

$$\therefore \text{No of solutions} = n = 4$$

$$\sum_{k=0}^{\infty} \left(\frac{1}{n^k}\right) = \sum_{k=0}^{\infty} \left(\frac{1}{4^k}\right)$$

$$= \frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots$$

$$= \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}$$

Q64: Let $A = [a_{ij}]$ be a real matrix of order 3×3 , such that $a_{i1} + a_{i2} + a_{i3} = 1$, for $i = 1, 2, 3$. Then, the sum of all the entries of the matrix A^3 is equal to:

- (A) 1
- (B) 3
- (C) 2
- (D) 9

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(A) 1

(B) 3

(C) 2

(D) 9

Solution:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{Let } x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$AX = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow AX = X$$

Replace X by AX

$$A^2 X = AX = X$$

Replace X by AX

$$A^3 X = AX = X$$

$$\text{Let } A^3 = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

$$A^3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 + x_3 \\ y_1 + y_2 + y_3 \\ z_1 + z_2 + z_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Sum of all the element = 3

Q65: Let a line $L : 2x + y = k, k > 0$ be a tangent to the hyperbola $x^2 - y^2 = 3$. If L is also a tangent to the parabola $y^2 = \alpha x$, then α is equal to :

- (A) 24
- (B) -12
- (C) -24
- (D) 12

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- (A) 24
- (B) -12
- (C) -24
- (D) 12

Solution:

Tangent to hyperbola of

Slope $m = -2$ (given)

$$y = -2x \pm \sqrt{3(3)}$$

$$(y = mx \pm \sqrt{a^2m^2 - b^2})$$

$$\Rightarrow y + 2x = \pm 3 \Rightarrow 2x + y = 3 (k > 0)$$

For parabola $y^2 = \alpha x$

$$y = mx + \frac{\alpha}{4m}$$

$$\Rightarrow y = -2x + \frac{\alpha}{-8}$$

$$\Rightarrow \frac{\alpha}{-8} = 3$$

$$\Rightarrow \alpha = -24.$$

Q66: If $\int_0^{100\pi} \frac{\sin^2 x}{e^{\left(\frac{x}{\pi} - \left[\frac{x}{\pi}\right]\right)}} dx = \frac{\alpha\pi^2}{1+4\pi^2}, \alpha \in R,$

where $[x]$ is the greatest integer less than or equal to x , then the value of α is:

- (A) $100(1 - e)$
- (B) $200(1 - e^{-1})$
- (C) $150(e^{-1} - 1)$
- (D) $50(e - 1)$

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(A) $100(1 - e)$

(B) $200(1 - e^{-1})$

(C) $150(e^{-1} - 1)$

(D) $50(e - 1)$

Solution:

$$\begin{aligned} I &= \int_0^{100\pi} \frac{\sin^2 x}{e^{\left\{\frac{x}{\pi}\right\}}} dx = 100 \int_0^{\pi} \frac{\sin^2 x}{e^{\frac{x}{\pi}}} dx \\ &= 100 \int_0^{\pi} e^{-\frac{x}{\pi}} \frac{(1 - \cos 2x)}{2} dx \\ &= 50 \left\{ \int_0^{\pi} e^{-\frac{x}{\pi}} dx - \int_0^{\pi} e^{-\frac{x}{\pi}} \cos 2x dx \right\} \\ I_1 &= \int_0^{\pi} e^{-\frac{x}{\pi}} dx = \left[-\pi e^{-\frac{x}{\pi}} \right]_0^{\pi} = \pi (1 - e^{-1}) \\ I_2 &= \int_0^{\pi} e^{-\frac{x}{\pi}} \cos 2x dx \\ &= -\pi e^{-\frac{x}{\pi}} \cos 2x \Big|_0^{\pi} - \int_0^{\pi} -\pi e^{-\frac{x}{\pi}} (-2 \sin 2x) dx \\ &= \pi (1 - e^{-1}) - 2\pi \int_0^{\pi} e^{-\frac{x}{\pi}} \sin 2x dx \\ &= \pi (1 - e^{-1}) - 2\pi \left\{ \left[-\pi e^{-\frac{x}{\pi}} \sin 2x \right]_0^{\pi} - \int_0^{\pi} \pi e^{-\frac{x}{\pi}} 2 \cos 2x dx \right\} \\ &= \pi (1 - e^{-1}) - 4\pi^2 I_2 \\ \Rightarrow I_2 &= \frac{\pi(1 - e^{-1})}{1 + 4\pi^2} \\ \therefore I &= 50 \left\{ \pi (1 - e^{-1}) - \frac{\pi(1 - e^{-1})}{1 + 4\pi^2} \right\} \\ &= \frac{200(1 - e^{-1})\pi^3}{1 + 4\pi^2} \end{aligned}$$

Q67: Four dice are thrown simultaneously and the numbers shown on these dice are recorded in 2×2 matrices. The probability that such formed matrices have all different entries and are non-singular, is:

(A) $\frac{23}{81}$

(B) $\frac{22}{81}$

(C) $\frac{45}{162}$

(D) $\frac{43}{162}$

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(C) $\frac{45}{162}$

(D) $\frac{43}{162}$

Solution:

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} |A| = ad - bc$$

Total case = 6^4

For non-singular matrix $|A| \neq 0 \Rightarrow ad - bc \neq 0$

$\Rightarrow ad \neq bc$

And a, b, c, d are all different numbers in the set $\{1, 2, 3, 4, 5, 6\}$

Now for $ad = bc$

(i) $6 \times 1 = 2 \times 3$

$\Rightarrow a = 6, b = 2, c = 3, d = 1$
 $or a = 1, b = 2, c = 3, d = 6$ } 8 such cases

(ii) $6 \times 2 = 3 \times 4$

$\Rightarrow a = 6, b = 2, c = 3, d = 2$
 $or a = 1, b = 3, c = 4, d = 6$ } 8 such cases

favourable cases

$$= {}^6C_4 (4!) - 16$$

required probability

$$= \frac{{}^6C_4 (4!) - 16}{6^4} = \frac{43}{162}$$

Q68: Let three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{a} \times \vec{b} = \vec{c}$, $\vec{b} \times \vec{c} = \vec{a}$ and $|\vec{a}| = 2$. Then which one of the following is not true?

(A) Projection of $\vec{a}$ on $(\vec{b} \times \vec{c})$ is 2

(B) $|3\vec{a} + \vec{b} - 2\vec{c}|^2 = 51$

(C) $[\vec{a} \vec{b} \vec{c}] + [\vec{c} \vec{a} \vec{b}] = 8$

(D) $\vec{a} \times ((\vec{b} + \vec{c}) \times (\vec{b} - \vec{c})) = \vec{0}$

Q68: Let three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ be such that $\vec{a} \times \vec{b} = \vec{c}$, $\vec{b} \times \vec{c} = \vec{a}$ and $|\vec{a}| = 2$. Then which one of the following is not true?

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(D) $\vec{a} \times ((\vec{b} + \vec{c}) \times (\vec{b} - \vec{c})) = \vec{0}$

Solution:

Projection of $\vec{a}$ on $\vec{b} \times \vec{c}$

$$= \frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{|\vec{b} \times \vec{c}|} = \frac{\vec{a} \cdot \vec{a}}{|\vec{a}|} = |\vec{a}| = 2$$

$$(2) \vec{a} \times \vec{b} = \vec{c} \text{ and } \vec{b} \times \vec{c} = \vec{a}$$

$\Rightarrow \vec{a}, \vec{b}, \vec{c}$ are mutually $\perp$ vectors.

$$|\vec{a} \times \vec{b}| = |\vec{c}| \Rightarrow |\vec{a}| |\vec{b}| = |\vec{c}| \Rightarrow |\vec{b}| = |\vec{c}| / 2$$

$$\text{Also, } |\vec{b} \times \vec{c}| = |\vec{a}| \Rightarrow |\vec{b}| |\vec{c}| = 2 \Rightarrow |\vec{c}| = 2 \text{ \& } |\vec{b}| = 1$$

$$|3\vec{a} + \vec{b} - 2\vec{c}|^2 = (3\vec{a} + \vec{b} - 2\vec{c}) \cdot (3\vec{a} + \vec{b} - 2\vec{c})$$

$$= 9|\vec{a}|^2 + |\vec{b}|^2 + 4|\vec{c}|^2$$

$$= (9 \times 4) + 1 + (4 \times 4)$$

$$= 36 + 1 + 16 = 53$$

$$(3) [\vec{a} \vec{b} \vec{c}] + [\vec{c} \vec{a} \vec{b}] = 2 [\vec{a} \vec{b} \vec{c}] = 2\vec{a} \cdot (\vec{b} \times \vec{c})$$

$$= 2\vec{a} \cdot \vec{a} = 2|\vec{a}|^2 = 8$$

$$(4) \vec{a} \times ((\vec{b} + \vec{c}) \times (\vec{b} - \vec{c}))$$

$$= \vec{a}(-\vec{b} \times \vec{c} + \vec{c} \times \vec{b}) = -2(\vec{a} \times (\vec{b} \times \vec{c}))$$

$$= -2(\vec{a} \times \vec{a}) = \vec{0}$$

Q69: Let S_n denote the sum of first n-terms of an arithmetic progression. If $S_{10} = 530, S_5 = 140$, then $S_{20} - S_6$ is equal to:

- (A) 1852
- (B) 1842
- (C) 1872
- (D) 1862

Q69: Let S_n denote the sum of first n-terms of an arithmetic progression. If $S_{10} = 530$, $S_5 = 140$, then $S_{20} - S_6$ is equal to:

- (A) 1852
- (B) 1842
- (C) 1872
- (D) 1862

Solution:

$$S_{10} = 530 \Rightarrow \frac{10}{2}\{2a + 9d\} = 530$$

$$\Rightarrow 2a + 9d = 106 \dots (1)$$

$$\text{and } S_5 = 140 \Rightarrow \frac{5}{2}\{2a + 4d\} = 140$$

$$\Rightarrow 2a + 4d = 56 \dots (2)$$

$$\Rightarrow 5d = 50 \Rightarrow d = 10 \Rightarrow a = 8$$

$$\text{Now, } S_{20} - S_6 = \frac{20}{2}\{2a + 19d\} - \frac{6}{2}\{2a + 5d\}$$

$$= 14a + 175d$$

$$= (14 \times 8) + (175 \times 10)$$

$$= 1862$$

Q70: Which of the following Boolean expressions is not a tautology?

(A) $(\sim p \Rightarrow q) \vee (\sim q \Rightarrow p)$

(B) $(p \Rightarrow q) \vee (\sim q \Rightarrow p)$

(C) $(p \Rightarrow q) \vee (q \Rightarrow \sim p)$

(D) $(p \Rightarrow \sim q) \vee (\sim q \Rightarrow p)$

Q70: Which of the following Boolean expressions is not a tautology?

(A) $(\sim p \Rightarrow q) \vee (\sim q \Rightarrow p)$

(B) $(p \Rightarrow q) \vee (\sim q \Rightarrow p)$

(C) $(p \Rightarrow q) \vee (q \Rightarrow \sim p)$

(D) $(p \Rightarrow \sim q) \vee (\sim q \Rightarrow p)$

Solution:

$$\begin{aligned} & (\sim p \rightarrow q) \vee (\sim q \rightarrow p) \\ &= (p \vee q) \vee (q \vee p) \\ &= (p \vee q) \vee (q \vee p) \\ &= p \vee q \end{aligned}$$

which is not a tautology.

$$\begin{aligned} (2) \quad & (q \rightarrow p) \vee (\sim q \rightarrow p) \\ &= (\sim q \vee p) \vee (q \vee p) \\ &= (\sim q \vee q) \vee p \\ &= t \vee p = t \end{aligned}$$

$$\begin{aligned} (3) \quad & (p \rightarrow q) \vee (\sim q \rightarrow p) \\ &= (\sim p \vee q) \vee (q \vee p) \\ &= (\sim p \vee p) \vee q \\ &= t \vee q = t \end{aligned}$$

$$\begin{aligned} (4) \quad & (p \rightarrow \sim q) \vee (\sim q \rightarrow p) \\ &= (\sim p \vee \sim q) \vee (q \vee p) \\ &= (\sim p \vee q) \vee (\sim q \vee q) \\ &= t \vee t = t \end{aligned}$$

Q71: Let the circle $36x^2 + 36y^2 - 108x + 120y + C = 0$ be such that it neither intersects nor touches the co-ordinate axes. If the point of intersection of the lines, $x - 2y = 4$ and $2x - y = 5$ lies inside the circle S, then :

(A) $100 < C < 156$

(B) $\frac{25}{9} < C < \frac{13}{3}$

(C) $81 < C < 156$

(D) $100 < C < 165$

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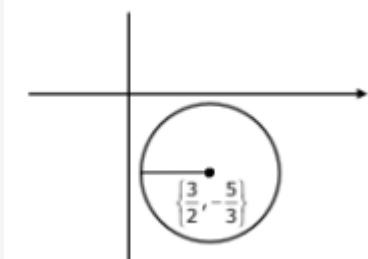
Solution:

$$S : 36x^2 + 36y^2 - 108x + 120y + C = 0$$

$$\Rightarrow x^2 + y^2 - 3x + \frac{10}{3}y + \frac{C}{36} = 0$$

$$\text{Centre} \equiv (-g, -f) \equiv \left(\frac{3}{2}, -\frac{10}{6}\right)$$

$$\text{radius} = r = \sqrt{\frac{9}{4} + \frac{100}{36} - \frac{C}{36}}$$



Now,

$$\Rightarrow r < \frac{3}{2}$$

$$\Rightarrow \frac{9}{4} + \frac{100}{36} - \frac{C}{36} < \frac{9}{4}$$

Now point of intersection of $x - 2y = 4$ and $2x - y = 5$ is $(2, -1)$, which lies inside the circle S.

$$\therefore S(2, -1) < 0$$

$$\Rightarrow (2)^2 + (-1)^2 - 3(2) + \frac{10}{3}(-1) + \frac{C}{36} < 0$$

$$\Rightarrow 4 + 1 - 6 - \frac{10}{3} + \frac{C}{36} < 0$$

$$C < 156$$

From (1) & (2)

$$100 < C < 156$$

Q72: The number of solutions of $\sin^7 x + \cos^7 x = 1, x \in [0, 4\pi]$ is equal to:

- (A) 5
- (B) 9
- (C) 11
- (D) 7

Q72: The number of solutions of $\sin^7 x + \cos^7 x = 1, x \in [0, 4\pi]$ is equal to:

(A) 5

(B) 9

(C) 11

(D) 7

Solution:

$$\sin^7 x \leq \sin^2 x \leq 1 \quad \dots (1)$$

$$\text{and } \cos^7 x \leq \cos^2 x \leq 1 \quad \dots (2)$$

$$\text{also } \sin^2 x + \cos^2 x = 1$$

$\Rightarrow$ equality must hold for (1) & (2)

$$\Rightarrow \sin^7 x = \sin^2 x \text{ \& } \cos^7 x = \cos^2 x$$

$$\Rightarrow \sin x = 0 \text{ \& } \sin x = 1 \text{ or}$$

$$\cos x = 0 \text{ \& } \sin x = 1$$

$$\Rightarrow x = 0, 2\pi, 4\pi, \frac{\pi}{2}, \frac{5\pi}{2}$$

$\Rightarrow$ 5 solutions

Q73: Let $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$. Let E_2 be another ellipse such that it touches the end points of major axis of E_1 and the foci E_2 are the end points of minor axis of E_1 . If E_1 and E_2 have same eccentricities, then its value is :

(A) $\frac{-1+\sqrt{3}}{2}$

(B) $\frac{-1+\sqrt{6}}{2}$

(C) $\frac{-1+\sqrt{5}}{2}$

(D) $\frac{-1+\sqrt{8}}{2}$

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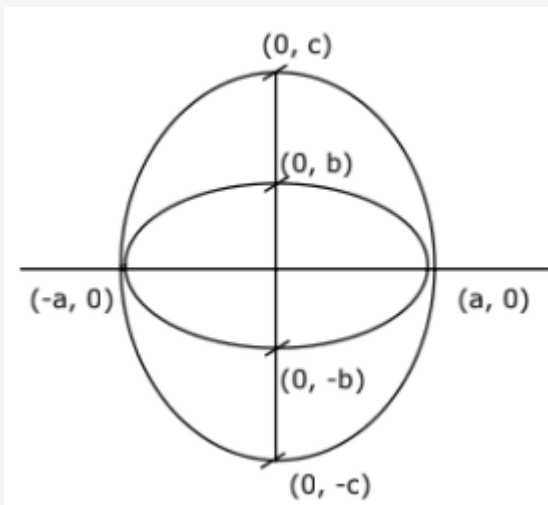
(A) $\frac{-1+\sqrt{3}}{2}$

(B) $\frac{-1+\sqrt{6}}{2}$

(C) $\frac{-1+\sqrt{5}}{2}$

(D) $\frac{-1+\sqrt{8}}{2}$

Solution:



$$e^2 = 1 - \frac{b^2}{a^2} \quad \dots (1)$$

$$e^2 = 1 - \frac{a^2}{c^2} \quad \dots (2)$$

$$\Rightarrow \frac{b^2}{a^2} = \frac{a^2}{c^2}$$

$$\Rightarrow c^2 = \frac{a^4}{b^2} \Rightarrow c = \frac{a^2}{b}$$

$$\text{Also } b = ce$$

$$\therefore \frac{b}{e} = \frac{a^2}{b} \Rightarrow \frac{b^2}{a^2} = e$$

$$e^2 = 1 - \frac{b^2}{a^2} \Rightarrow e^2 + e - 1 = 0$$

$$e = \frac{-1 + \sqrt{5}}{2}$$

Q74: Let L be the line of intersection of planes $\vec{r} \cdot (\hat{i} - \hat{j} + 2\hat{k}) = 2$ and

$\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) = 2$. If $P(\alpha, \beta, \gamma)$ is the foot of perpendicular on L from the point (1, 2, 0), then the value of $35(\alpha + \beta + \gamma)$ is equal to :

(A) 134

(B) 119

(C) 143

(D) 101

Q74: Let L be the line of intersection of planes $\vec{r} \cdot (\hat{i} - \hat{j} + 2\hat{k}) = 2$ and

$\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) = 2$. If $P(\alpha, \beta, \gamma)$ is the foot of perpendicular on L from the point (1, 2, 0), then the value of $35(\alpha + \beta + \gamma)$ is equal to :

(A) 134

(B) 119

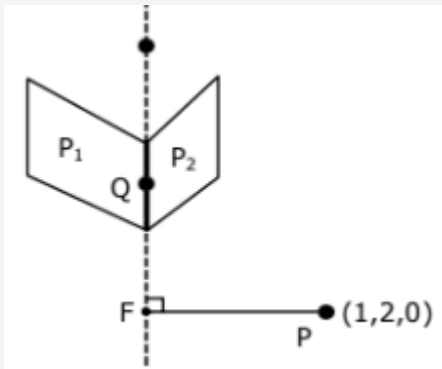
(C) 143

(D) 101

Solution:

$$P_1 : x - y + 2z = 2$$

$$P_2 : 2x + y - z = 2$$



Let line of intersection of planes P_1 and P_2 cuts xy plane in point Q.

$\Rightarrow$ z - coordinate of point Q is zero

$$\Rightarrow \left. \begin{array}{l} x - y = 2 \\ 2x + y = 2 \end{array} \right\} \Rightarrow x = \frac{4}{3}, y = \frac{-2}{3}$$

$$\Rightarrow Q \left(\frac{4}{3}, \frac{-2}{3}, 0 \right)$$

Vector parallel to the line of intersection

$$\vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix} = -\hat{i} + 5\hat{j} + 3\hat{k}$$

Equation of line of intersection

$$\frac{x - \frac{4}{3}}{-1} = \frac{y + \frac{2}{3}}{5} = \frac{z - 0}{3} = \lambda \text{ (say)}$$

Let coordinates of foot of perpendicular be

$$F \left(-\lambda + \frac{4}{3}, 5\lambda - \frac{2}{3}, 3\lambda \right)$$

$$\overrightarrow{PF} = \left(-\lambda + \frac{1}{3}\right) \hat{i} + \left(5\lambda - \frac{8}{3}\right) \hat{j} + (3\lambda) \hat{k}$$

$$\overrightarrow{PF} \cdot \vec{a} = 0$$

$$\Rightarrow \lambda - \frac{1}{3} + 25\lambda - \frac{40}{3} + 9\lambda = 0$$

$$\Rightarrow 35\lambda = \frac{41}{3} \Rightarrow \lambda = \frac{41}{105}$$

$$\text{Now, } \alpha = -\lambda + \frac{4}{3}, \beta = 5\lambda - \frac{2}{3}, \gamma = 3\lambda$$

$$\Rightarrow \alpha + \beta + \gamma = 7\lambda + \frac{2}{3}$$

$$= 7\left(\frac{41}{105}\right) + \frac{2}{3}$$

$$= \frac{51}{15}$$

$$\Rightarrow 35(\alpha + \beta + \gamma) = \frac{51}{15} \times 35 = 119$$

Q75: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as
$$f(x) = \begin{cases} \frac{x^3}{(1-\cos 2x)^2} \log_e \left(\frac{1+2xe^{-2x}}{(1-xe^{-x})^2} \right) & x \neq 0 \\ \alpha & x = 0 \end{cases}$$

If f is continuous at $x = 0$, then α is equal to :

(A) 1

(B) 0

(C) 3

(D) 2

Q75: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} \frac{x^3}{(1-\cos 2x)^2} \log_e \left(\frac{1+2xe^{-2x}}{(1-xe^{-x})^2} \right) & x \neq 0 \\ \alpha & x = 0 \end{cases}$

If f is continuous at $x = 0$, then α is equal to :

(A) 1

(B) 0

(C) 3

(D) 2

Solution:

For continuity

$$\lim_{x \rightarrow 0} \frac{x^3}{4\sin^4 x} (\ln(1 + 2xe^{-2x}) - 2\ln(1 - xe^{-x}))$$

$$= \alpha$$

$$\lim_{x \rightarrow 0} \frac{1}{4x} [2xe^{-2x} + 2xe^{-x}] = \alpha$$

$$= \frac{1}{4}(4) = \alpha = 1$$

Q76: Let a vector $\vec{a}$ be coplanar with vectors $\vec{b} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$. If $\vec{a}$ is perpendicular to $\vec{d} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, and $|\vec{a}| = \sqrt{10}$. Then a possible value of $[\vec{a} \ \vec{b} \ \vec{c}] + [\vec{a} \ \vec{d} \ \vec{b}] + [\vec{a} \ \vec{c} \ \vec{d}]$ is equal to :

- (A) -40
- (B) -42
- (C) -29
- (D) -38

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(A) -40

(B) -42

(C) -29

(D) -38

Solution:

$$\vec{a} = \lambda \vec{b} + \mu \vec{c} = \hat{i}(2\lambda + \mu) + \hat{j}(\lambda - \mu) + \hat{k}(\lambda + \mu)$$

$$\vec{a} \cdot \vec{d} = 0 = 3(2\lambda + \mu) + 2(\lambda - \mu) + 6(\lambda + \mu)$$

$$\Rightarrow 14\lambda + 7\mu = 0 \Rightarrow \mu = -2\lambda$$

$$\Rightarrow \vec{a} = (0)\hat{i} - 3\lambda\hat{j} + (-\lambda)\hat{k}$$

$$\Rightarrow |\vec{a}| = \sqrt{10}|\lambda| = \sqrt{10} \Rightarrow |\lambda| = 1$$

$$\Rightarrow \lambda = 1 \text{ or } -1$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0 \text{ (as vectors are coplanar)}$$

$$[\vec{a} \ \vec{b} \ \vec{c}] + [\vec{a} \ \vec{b} \ \vec{d}] + [\vec{a} \ \vec{c} \ \vec{d}] = [\vec{a} \ \vec{b} + \vec{c} \ \vec{d}]$$

$$= \begin{vmatrix} 0 & -3\lambda & \lambda \\ 3 & 0 & 2 \\ 3 & 2 & 6 \end{vmatrix}$$

$$= 3\lambda(12) + \lambda(6) = 42\lambda = -42$$

Q77: Let $y = y(x)$ be the solution of the differential equation $\operatorname{cosec}^2 x \, dy + 2dx = (1 + y \cos 2x) \operatorname{cosec}^2 x \, dx$, with $y\left(\frac{\pi}{4}\right) = 0$. Then, the value of $(y(0) + 1)^2$ is equal to :

(A) $e^{\frac{1}{2}}$

(B) $e^{-\frac{1}{2}}$

(C) e^{-1}

(D) e

Q77: Let $y = y(x)$ be the solution of the differential equation $\operatorname{cosec}^2 x \, dy + 2dx = (1 + y \cos 2x) \operatorname{cosec}^2 x \, dx$, with $y\left(\frac{\pi}{4}\right) = 0$. Then, the value of $(y(0) + 1)^2$ is equal to :

(A) $e^{\frac{1}{2}}$

(B) $e^{-\frac{1}{2}}$

(C) e^{-1}

(D) e

Solution:

$$\frac{dy}{dx} + 2\sin^2 x = 1 + y \cos 2x$$

$$\Rightarrow \frac{dy}{dx} + (-\cos 2x) y = \cos 2x$$

$$I.F. = e^{\int -\cos 2x dx} = e^{-\frac{\sin 2x}{2}}$$

Solution of D.E.

$$y \cdot e^{\frac{-\sin 2x}{2}} = \int (\cos 2x) e^{\frac{-\sin 2x}{2}} dx + c$$

Given

$$y\left(\frac{\pi}{4}\right) = 0$$

$$\Rightarrow 0 = -e^{-\frac{1}{2}} + c \Rightarrow c = e^{-\frac{1}{2}}$$

at $x = 0$

$$y = -1 + e^{-\frac{1}{2}}$$

$$\Rightarrow y(0) = -1 + e^{-\frac{1}{2}}$$

$$\Rightarrow (y(0) + 1)^2 = e^{-1}$$

Q78: The value of λ and μ such that the system of equations

$x + y + z = 6$, $3x + 5y + 5z = 26$, $x + 2y + \lambda z = \mu$ has no solution, are :

(A) $\lambda = 3, \mu \neq 10$

(B) $\lambda \neq 2, \mu = 10$

(C) $\lambda = 3, \mu = 5$

(D) $\lambda = 2, \mu \neq 10$

Q78: The value of λ and μ such that the system of equations

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(A) $\lambda = 3, \mu \neq 10$

(B) $\lambda \neq 2, \mu = 10$

(C) $\lambda = 3, \mu = 5$

(D) $\lambda = 2, \mu \neq 10$

Solution:

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 3 & 5 & 5 \\ 1 & 2 & \lambda \end{vmatrix} = 0$$

$$(5\lambda - 10) - 1(3\lambda - 5) + (6 - 5) = 0$$

$$2\lambda - 10 + 5 + 1 = 0$$

$\lambda = 2$ only one option satisfy

Atleast one IF $D_1, D_2, D_3 \neq 0$

Q79: If the domain of the function $f(x) = \frac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\frac{2x-1}{2}\right)}}$ is the interval $(\alpha, \beta]$, then $\alpha + \beta$

is equal to :

(A) 2

(B) $\frac{3}{2}$

(C) $\frac{1}{2}$

(D) 1

Q79: If the domain of the function $f(x) = \frac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\frac{2x-1}{2}\right)}}$ is the interval $(\alpha, \beta]$, then $\alpha + \beta$

is equal to :

(A) 2

(B) $\frac{3}{2}$

(C) $\frac{1}{2}$

(D) 1

Solution:

$$0 \leq x^2 - x + 1 \leq 1$$

$$\Rightarrow x^2 - x \leq 0$$

$$\Rightarrow x \in [0, 1]$$

$$\text{Also, } 0 < \sin^{-1} \left(\frac{2x-1}{2} \right) \leq \frac{\pi}{2}$$

$$\Rightarrow 0 < \frac{2x-1}{2} \leq 1$$

$$\Rightarrow 0 < 2x - 1 \leq 2$$

$$1 < 2x \leq 3$$

$$\frac{1}{2} < x \leq \frac{3}{2}$$

Taking intersection

$$x \in \left(\frac{1}{2}, 1 \right]$$

$$\Rightarrow \alpha = \frac{1}{2}, \beta = 1$$

$$\Rightarrow \alpha = \frac{1}{2}, \beta = 1$$

Q80: If the shortest distance between the straight lines $3(x - 1) = 6(y - 2) = 2(z - 1)$ and $4(x - 2) = 2(y - \lambda) = (z - 3)$, $\lambda \in \mathbb{R}$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to :

(A) -1

(B) 2

(C) 3

(D) 5

Q80: If the shortest distance between the straight lines $3(x - 1) = 6(y - 2) = 2(z - 1)$ and $4(x - 2) = 2(y - \lambda) = (z - 3)$, $\lambda \in \mathbb{R}$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to :

(A) -1

(B) 2

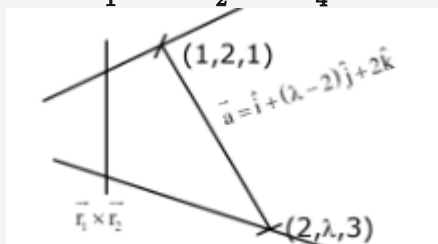
(C) 3

(D) 5

Solution:

$$L_1 : \frac{(x-1)}{2} = \frac{(y-2)}{1} = \frac{(z-1)}{3}, \quad \vec{r}_1 = 2\hat{i} + \hat{j} + 3\hat{k}$$

$$L_2 : \frac{(x-2)}{1} = \frac{(y-\lambda)}{2} = \frac{(z-3)}{4}, \quad \vec{r}_2 = \hat{i} + 2\hat{j} + 4\hat{k}$$



Shortest distance = Projection of $\vec{a}$ on $\vec{r}_1 \times \vec{r}_2$

$$= \frac{|\vec{a} \cdot (\vec{r}_1 \times \vec{r}_2)|}{|\vec{r}_1 \times \vec{r}_2|}$$

$$|\vec{a} \cdot (\vec{r}_1 \times \vec{r}_2)| = \begin{vmatrix} 1 & \lambda - 2 & 2 \\ 2 & 1 & 3 \\ 1 & 2 & 4 \end{vmatrix} = |14 - 5\lambda|$$

$$|\vec{r}_1 \times \vec{r}_2| = \sqrt{38}$$

$$\therefore |14 - 5\lambda| = 1$$

$$\Rightarrow 14 - 5\lambda = 1 \text{ or } 14 - 5\lambda = -1$$

$$\Rightarrow \lambda = \frac{13}{5} \text{ or } 3$$

$\therefore$ Integral value of $\lambda = 3$

Q81: The number of elements in the set

$\{n \in \{1, 2, 3, \dots, 100\} \mid (11)^n > (10)^n + (9)^n\}$ is _____. is

Q81: The number of elements in the set

$\{n \in \{1, 2, 3, \dots, 100\} \mid (11)^n > (10)^n + (9)^n\}$ is _____. is

96.00

Solution:

$$11^n > 10^n + 9^n$$

$$\Rightarrow 11^n - 9^n > 10^n$$

$$\Rightarrow (10 + 1)^n - (10 - 1)^n > 10^n$$

$$\Rightarrow \{ {}^nC_1 \cdot 10^{n-1} + {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \} > 10^n$$

$$\Rightarrow 2n \cdot 10^{n-1} + 2 \{ {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \} > 10^n \quad \dots (1)$$

For $n = 6, 7, 8, \dots, 100$

$$\Rightarrow 2n \cdot 10^{n-1} + \{ {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \} > 10^n$$

$$\Rightarrow 11^n - 9^n > 10^n \text{ For } n = 5, 6, 7, \dots, 100$$

For $n = 4$, Inequality (1) is not satisfied

$\Rightarrow$ Inequality does not hold good for $N = 1, 2, 3, 4$

So, required number of elements

= 96

Q82: Let $y = y(x)$ be the solution of the differential equation

$$\left((x+2) e^{\left(\frac{y+1}{x+2}\right)} + (y+1) \right) dx = (x+2) dy, y(1) = 1.$$

If the domain of $y = y(x)$ is an open interval (α, β) , then $|\alpha + \beta|$ is equal to _____.

Q82: Let $y = y(x)$ be the solution of the differential equation

$$\left((x+2) e^{\left(\frac{y+1}{x+2}\right)} + (y+1) \right) dx = (x+2) dy, y(1) = 1.$$

If the domain of $y = y(x)$ is an open interval (α, β) , then $|\alpha + \beta|$ is equal to _____.

4.00

Solution:

$$y + 1 = Y \Rightarrow dy = dY$$

$$x + 2 = X \Rightarrow dx = dX$$

$$\Rightarrow \left(X e^{\frac{Y}{X}} + Y \right) dX = X dY$$

$$\Rightarrow X dY - Y dX = X e^{Y/X} dX$$

$$\Rightarrow d\left(\frac{Y}{X}\right) e^{-\frac{Y}{X}} = \frac{dX}{X}$$

$$-e^{-\frac{Y}{X}} = \ln|X| + c$$

$$(3, 2) \rightarrow -e^{-\frac{2}{3}} = \ln|3| + c$$

$$-e^{-\frac{Y}{X}} = \ln|X| - e^{-\frac{2}{3}} - \ln 3$$

$$e^{-\frac{Y}{X}} = e^{\frac{2}{3}} + \ln 3 - \ln|X| > 0$$

$$\ln|X| < e^{\frac{2}{3}} + \ln 3 \rightarrow \lambda$$

$$|x + 2| < e^\lambda$$

$$-e^\lambda < x + 2 < e^\lambda$$

$$-e^\lambda - 2 < x < e^\lambda - 2$$

$$\alpha + \beta = -4 \Rightarrow |\alpha + \beta| = 4$$

Q83: Let $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then the number of bijective functions $f : A \rightarrow A$ such that $f(1) + f(2) = 3 - f(3)$ is equal to _____.

Q83: Let $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then the number of bijective functions $f : A \rightarrow A$ such that $f(1) + f(2) = 3 - f(3)$ is equal to _____.

720.00

Solution:

$$f(1) + f(2) = 3 - f(3)$$

$$\Rightarrow f(1) + f(2) + f(3) = 3$$

The only possibility is: $0 + 1 + 2 = 3$

$\Rightarrow$ Elements 1, 2, 3 in the domain can be mapped with 0, 1, 2 only.

So number of bijective functions.

$$= 3! \times 5! = 720$$

Q84: If the constant term, in binomial expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10}$ is 180, then r is equal to

Q84: If the constant term, in binomial expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10}$ is 180, then r is equal to

8.00

Solution:

$$\left(2x^r + \frac{1}{x^2}\right)^{10}$$

$$\text{General term} = {}^{10}C_R (2x^r)^{10-R} x^{-2R}$$

$$\Rightarrow 2^{10-R} {}^{10}C_R = 180 \quad \dots (1)$$

$$\& (10 - R)r - 2R = 0$$

$$r = \frac{2R}{10-R}$$

$$r = \frac{2(R-10)}{10-R} + \frac{20}{10-R}$$

$$\Rightarrow r = -2 + \frac{20}{10-R} \dots (2)$$

$R = 8$ or 5 reject equation (1) not satisfied

At $R = 8$

$$2^{10-R} {}^{10}C_R = 180 \Rightarrow r = 8$$

Q85: Let $r : R \rightarrow R$ be function defined as $f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & \text{if } |x| \leq 2 \\ 0 & \text{if } |x| > 2 \end{cases}$

Let $g : R \rightarrow R$ be given by $g(x) = f(x+2) - f(x-2)$. If n and m denote the number of points in R where g is not continuous and not differentiable, respectively, then $n + m$ is equal to

Q85: Let $r : \mathbb{R} \rightarrow \mathbb{R}$ be function defined as $f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & \text{if } |x| \leq 2 \\ 0 & \text{if } |x| > 2 \end{cases}$

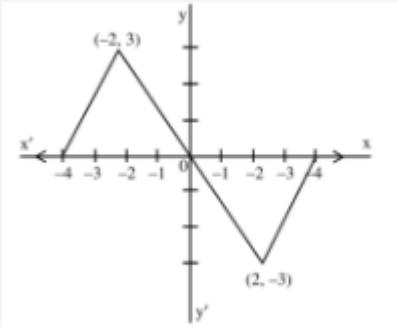
Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be given by $g(x) = f(x+2) - f(x-2)$. If n and m denote the number of points in $\mathbb{R}$ where g is not continuous and not differentiable, respectively, then $n + m$ is equal to

Solution:

$$f(x-2) = \begin{cases} \frac{3x}{2} & -4 \leq x \leq -2 \\ -\frac{3x}{2} & -2 < x \leq 0 \\ 0 & x \in (-\infty, -4) \cup (0, \infty) \end{cases}$$

$$f(x+2) = \begin{cases} \frac{3x}{2} + 6 & 0 \leq x \leq 2 \\ -\frac{3x}{2} + 6 & 2 < x \leq 4 \\ 0 & x \in (-\infty, 0) \cup (4, \infty) \end{cases}$$

$$g(x) = f(x+2) - f(x-2) = \begin{cases} \frac{3x}{2} + 6 & -4 \leq x \leq -2 \\ -\frac{3x}{2} & -2 < x < 2 \\ \frac{3x}{2} - 6 & 2 \leq x \leq 4 \\ 0 & x \in (-\infty, -4) \cup (4, +\infty) \end{cases}$$



$$n = 0$$

$$m = 4 \Rightarrow (n + m = 4)$$

Q86: Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then the number of 3×3 matrices B with entries from the set $\{1, 2, 3, 4, 5\}$ and satisfying $AB = BA$ is _____.

Q86: Let $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Then the number of 3×3 matrices B with entries from the set $\{1, 2, 3, 4, 5\}$ and satisfying $AB = BA$ is _____.

Solution:

$$\text{Let Matrix } B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\therefore AB = BA$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix} = \begin{bmatrix} b & a & c \\ e & d & f \\ h & g & i \end{bmatrix}$$

$$\Rightarrow d = b, e = a, f = c, g = h$$

$$\therefore \text{Matrix } B = \begin{bmatrix} a & b & c \\ b & a & c \\ g & g & i \end{bmatrix}$$

No. of ways of selecting $a, b, c, g,$

$$i = 5 \times 5 \times 5 \times 5 \times 5$$

$$= 5^5 = 3125$$

$$\therefore \text{No. of Matrices } B = 3125$$

Q87: Consider the following frequency distribution :

Class:	0-6	6-12	12-18	18-24	24-30
Frequency :	a	b	12	9	5

If mean = $\frac{309}{22}$ and median = 14, then value $(a - b)^2$ is equal to _____.

Q87: Consider the following frequency distribution :

Class:	0-6	6-12	12-18	18-24	24-30
Frequency :	a	b	12	9	5

If mean = $\frac{309}{22}$ and median = 14, then value $(a - b)^2$ is equal to _____.

4.00

Solution:

Class:	Frequency	X_i	$F_i x_i$
0-6	a	3	$3a$
6-12	b	9	$9b$
12-18	12	15	180
18-24	9	21	189
24-30	5	27	135
	$N = (26 + a + b)$		$(504 + 3a + 9b)$

$$\text{Mean} = \frac{3a+9b+180+189+135}{a+b+26} = \frac{309}{22}$$

$$\Rightarrow 66a + 198b + 11088 = 309a + 309b + 8034$$

$$\Rightarrow 243a + 111b = 3054$$

$$\Rightarrow 81a + 37b = 1018 \quad \dots (1)$$

$$\text{Now, Median} = 12 + \frac{\frac{a+b+26}{2} - (a+b)}{12} \times 6 = 14$$

$$\Rightarrow \frac{13}{2} - \left(\frac{a+b}{4} \right) = 2$$

$$\Rightarrow \frac{a+b}{4} = \frac{9}{2}$$

$$\Rightarrow a + b = 18 \rightarrow (2)$$

From equation (1) & (2)

$$a = 8, b = 10$$

$$\therefore (a - b)^2 = (8 - 10)^2$$

Q88: If the digits are not allowed to repeat in any number formed by using the digits 0, 2, 4, 6, 8, then the number of all numbers greater than 10,000 is equal to _____.

Q88: If the digits are not allowed to repeat in any number formed by using the digits 0, 2, 4, 6, 8, then the number of all numbers greater than 10,000 is equal to _____.

96.00

Solution:

0, 2, 4, 6, 8				
2, 4, 6, 8				
4	4	3	2	1

$\Rightarrow 4 \times 4 \times 3 \times 2 = 96$

Q89: The sum of all the elements in the set $\{n \in \{1, 2, \dots, 100\} \mid H.F.C \text{ of } n \text{ and } 2040 \text{ is } 1\}$ is equal to _____.

Q89: The sum of all the elements in the set $\{n \in \{1, 2, \dots, 100\} \mid H.F.C \text{ of } n \text{ and } 2040 \text{ is } 1\}$ is equal to _____.

1251.00

Solution:

$$2040 = 2^3 \times 3 \times 5 \times 17$$

n should not be multiple of 2, 3, 5 and 17.

$$\begin{aligned} \text{Sum of all } n &= (1 + 3 + 5 \dots + 99) - (3 + 9 + 15 + 21 + \dots + 99) - (5 + 25 + 35 + 55 + 65 + 85 + 95) - (17) \\ &= 2500 - \frac{17}{2}(3 + 99) - 365 - 17 \\ &= 2500 - 867 - 365 - 17 = 1251 \end{aligned}$$

Q90: The area (in sq. units) of the region bounded by the curves

$x^2 + 2y - 1 = 0$, $y^2 + 4x - 4 = 0$ and $y^2 - 4x - 4 = 0$, in the upper half plane is

_____.

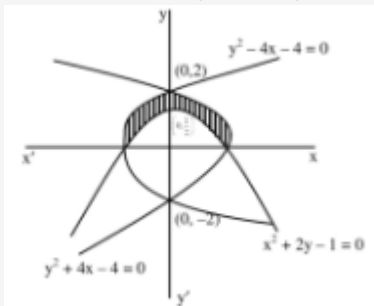
Q90: The area (in sq. units) of the region bounded by the curves

$x^2 + 2y - 1 = 0$, $y^2 + 4x - 4 = 0$ and $y^2 - 4x - 4 = 0$, in the upper half plane is

2.00

Solution:

Required Area (shaded)



$$= 2 \left[\int_0^2 \left(\frac{4-y^2}{4} \right) dy - \int_0^1 \left(\frac{1-x^2}{2} \right) dx \right]$$

$$= 2 \left[\frac{4}{3} - \frac{1}{3} \right] = (2)$$